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REGULAR CONTRIBUTION

Towards trustworthy foundation models: a systematic review of safety, evaluation, and defence mechanisms

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Abstract

Large Language Models (LLMs) and foundation models are increasingly deployed in security-critical and high-impact settings, including healthcare, cybersecurity, software engineering, and intelligent infrastructure. Their open-ended interfaces and multimodal capabilities create new attack surfaces, where prompt injection, jailbreaking, hallucination, adversarial fine-tuning, and cross-modal manipulation can compromise reliability, integrity, privacy, and user trust. This paper presents a systematic literature review of 94 peer-reviewed studies identified through database searching, backward reference snowballing, venue screening, and quality assessment within a January 2021–July 2026 search window. The review synthesises evidence on safety and robustness vulnerabilities, red-teaming practices, mitigation mechanisms, evaluation metrics, and unresolved research gaps for LLMs and related foundation-model systems. The findings show that LLM vulnerabilities are rarely isolated: prompt-level, model-level, data-level, and deployment-level risks often interact, particularly in high-stakes domains. Red-teaming has become more structured, but existing practices remain inconsistent across threat models, benchmarks, attack settings, and reporting standards. The reviewed defences include prompt hardening, input/output filtering, adversarial training, retrieval-augmented grounding, cross-model verification, gatekeeper models, and domain-specific safety layers; however, their effectiveness is difficult to compare because studies use heterogeneous metrics and evaluation protocols. Major gaps include fragmented benchmarking, limited multilingual and low-resource assessment, insufficient adaptive-adversary and long-term deployment studies, and weak integration between technical defences and governance mechanisms. The review indicates that trustworthy foundation-model deployment requires standardised threat models, reproducible red-teaming protocols, transparent safety metrics, and defence-in-depth strategies tailored to domain-specific security risks.

Keywords Large language models · Foundation models · Information security · Red-teaming · Adversarial robustness · Prompt injection · Jailbreaking · Multimodal security · Defence mechanisms

1 Introduction

Large Language Models (LLMs) and foundation models have moved beyond experimental artifacts and are now being rapidly deployed across a wide range of applications, including natural language understanding, multimodal reasoning, software engineering, healthcare analytics, and intelligent cyber-physical systems. Current developments demonstrate that LLMs can carry out sophisticated tasks, including auto-

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mated data visualisation generation [57], clinical decision-making and diagnostic reasoning [46], industrial-scale dialogue modelling [16], and real-time monitoring and fault recovery in intelligent infrastructures [25]. Although these changes demonstrate the transformational potential of LLM-based systems, they also raise serious concerns about safety, robustness, and resilience in harsh, high-risk scenarios.

Early studies in this field focused mainly on the efficiency of Architectural designs, scalability, and the feasibility of training Transformer-based models, aiming to minimise computational cost and expand availability [52]. As LLMs increasingly rely on prompt-based interaction and are deployed as user-facing, open-ended pipelines, new types of vulnerabilities have surfaced. These include prompt injection, jailbreak attacks, adversarial fine-tuning, and adversarial example misuse via indirect or multimodal inputs, which are not the same as the original adversarial examples studied in earlier machine learning literature [23]. These weaknesses cast doubt on the perception that good average-case behaviour reliably or safely reflects real-world system behaviour.

These vulnerabilities pose especially serious risks in applied and safety-critical fields. In the medical field, LLMs have been considered capable of performing clinical reasoning, generating various reports, and interacting with patients [46], though research consistently finds that they are ill-equipped to handle noisy inputs, produce hallucinated outputs, and are not as robust as they should be when subjected to adversarial manipulation. In software engineering and cybersecurity contexts, LLMs have been extended to vulnerability detection, program analysis, and the analysis of IoT devices and other insecure systems [4, 39]. Their vulnerability to adversarial prompts and shifting distributions raises questions about reliability and trustworthiness. In resource-intensive infrastructure and intelligent energy systems, where LLMs are increasingly integrated with real-time optimisation and control pipelines, robustness failures can cause cascading physical and economic impacts [25].

Socio-technical challenges extend beyond technical vulnerabilities. Prior research also highlights issues with bias dissemination, lack of transparency, and limited controllability in LLM-based interactive systems [48]. In order to address these problems, several solutions have been suggested, such as structured scenario engineering of foundation models [52], using knowledge injection and grounding methods to restrict the behaviour of models [11], and architecture-based dialogue that aims to enhance generalisation and control of heterogeneous tasks [16]. These mitigation strategies vary in effectiveness across fields, and they are not usually evaluated uniformly.

Although the current literature on the safety of LLMs, their robustness, and red-teaming is rapidly increasing, the studies remain disjointed. There is a lack of literature that provides a comprehensive view of the field because most researchers focus on attack vectors in isolation, single defence mechanisms, or narrowly scoped classes of application scenarios. Evaluation practices are also quite heterogeneous, ranging from qualitative case-based analyses to task-specific quantitative measurements, especially in high-stakes areas such as healthcare, cybersecurity, and intelligent systems [23, 25, 46]. This discontinuity impedes comparability, reproducibility, and the determination of a systematic evidence gap.

Several peer-reviewed reviews have examined complementary aspects of LLM security and safety. These include the dual role of LLMs in security and privacy [98], safety assurance through verification and validation [35], the construction and application of LLMs for cybersecurity tasks [104], safeguarding and guardrail mechanisms [19], end-to-end safety pipelines covering attacks, defences, alignment, metrics, and guardrails [38], and the technical and operational characteristics of modern jailbreak attacks [107]. Table 1 positions the present SLR relative to these complementary reviews.

This work is not positioned as the first review of LLM security. Its distinct contribution is a quality-screened, cross-domain synthesis that connects reported vulnerabilities to the methods used to expose them, the controls proposed to mitigate them, the conditions under which those controls were evaluated, and the residual risks remaining after mitigation. The studies also capture emerging primary evidence on autonomous and context-aware red-teaming [30, 92], self-evolving attacks [53], multilingual prompt injection [81], cross-layer injection surfaces in LLM-powered web agents [108], and prompt injection in vision–language surgical decision support [105].

This paper addresses these issues by conducting a systematic literature review (SLR) of 94 research articles on the safety, robustness, and red-teaming of LLMs and associated AI systems. The synthesis of the 94 quality-screened primary studies generates the following new knowledge:

- A hierarchical, multi-label taxonomy, informed by the review and aligned with established frameworks, that relates six attack surfaces to application domains, compromised security boundaries, and defence checkpoints. The six surfaces are the training and data supply chain, model assets and deployment configuration, instruction and interaction, external context and tools, multimodal fusion, and agent or workflow orchestration.
- A cross-study explanation of LLM safety as a system property: prompt-, context-, model-, modality-, and workflow-level weaknesses interact, so apparently distinct attacks can form a cross-layer attack path.
- A quantitative finding that 48 of 94 studies (51.1%) address multiple vulnerability categories and 81 (86.2%)

Table 1 Positioning of the present SLR relative to representative reviews of LLM security and safety

Review	Principal scope	Relationship to the present SLR
Yao et al. [98]	Beneficial security applications, malicious uses, privacy risks, vulnerabilities, and defences	Provides a broad dual-use security–privacy perspective. The present SLR traces study-level relationships among attack surfaces, red-teaming, defences, evaluation settings, and residual risks
Huang et al. [35]	Evaluation, verification, runtime monitoring, regulation, and ethical use from a verification-and-validation perspective	Provides a lifecycle-assurance perspective. The present SLR adds an RQ-based synthesis of empirical evidence across application and deployment domains
Zhang et al. [103]	Construction of cybersecurity-oriented LLMs and their use in cybersecurity tasks	Primarily examines *LLMs for cybersecurity*. The present SLR examines the security and robustness of LLM systems across cybersecurity, healthcare, intelligent infrastructure, business, multilingual, and multimodal settings
Dong et al. [19]	Safeguards, guardrails, safety properties, attacks, and lifecycle considerations	Concentrates on safeguarding mechanisms. The present SLR additionally maps vulnerabilities to red-teaming methods, mitigation evidence, evaluation constraints, and domain-specific risks
Jalan et al. [38]	Attacks, defences, alignment, evaluation metrics, and guardrails in an end-to-end safety pipeline	Provides an integrated safety taxonomy. The present SLR contributes an explicit search and quality-assessment protocol and a domain-coded synthesis of 94 peer-reviewed primary studies
Zhou et al. [107]	Technical and operational taxonomies of modern LLM jailbreak attacks	Focuses specifically on jailbreak feasibility. The present SLR covers a broader range of prompt-, model-, data-, multimodal-, deployment-, and domain-level risks together with their corresponding defences and evaluation evidence
This SLR	94 peer-reviewed primary studies published from January 2021 to July 2026	An integrated synthesis of vulnerabilities, red-teaming, defence mechanisms, evaluation practices, high-stakes application risks, and actionable research directions.

use multiple red-teaming methods, whereas only 38 report multiple mitigation strategies and 15 report none. This reveals that broader attack discovery is not yet matched by equally comprehensive defence evidence.

– A model-to-application evidence map connecting evaluated model families to healthcare, cybersecurity and software engineering, business and finance, intelligent infrastructure, multilingual and social-media, multimodal, and agentic applications, thereby exposing domain-dependent failure consequences and validation requirements.
– An evidence-grounded research agenda linking the findings to standardised threat-model reporting, cross-layer and adaptive-adversary benchmarks, multilingual and high-stakes validation, reproducible pre- and post-defence measurement, and integration of technical controls with deployment governance.

The rest of this paper is structured as follows: The review background is provided in 2, and they are organised around system-level, adversarial threat, defensive measures, and application-specific safety. 3 outlines the methodology that was used in carrying out the SLR, and this is given in terms of search strategy, selection criteria, and data extraction procedures. 4 discusses the findings against the stipulated research questions and includes quantitative and qualitative evidence. Lastly, 5 concludes the review by discussing the implications of the results at the next level, identifying research gaps, and specifying directions for future work.

2 Literature background

This section provides only the concepts needed to interpret the review; Sect. 4 reserves detailed study-level evidence.

2.1 Core security concepts and attack surfaces

An *LLM vulnerability* is a weakness in the model, training data, interface, connected tools, or deployment configuration that can produce unsafe, incorrect, privacy-compromising, or adversary-controlled behaviour. A *jailbreak* bypasses behavioural safeguards through instruction manipulation, obfuscation, optimisation, or multi-turn interaction [21, 28, 53, 100]. *Prompt injection* instead manipulates the task or instruction hierarchy: it may be direct, semantic-layer, or indirect through retrieved documents, webpages, memory, or tool outputs [59, 104, 108]. A *multimodal attack* carries the malicious signal through images or cross-modal inputs, including safety-critical visual decision support [50, 105].

Red-teaming is the structured adversarial evaluation used to reveal these weaknesses under an explicit threat model. It ranges from manual prompt testing to automated and autonomous agents that adapt attacks over multiple turns [30, 92, 95]. A *defence mechanism* is any preventive, detective, corrective, or monitoring control—for example, input filtering, adversarial training, grounded retrieval, response validation, or human escalation. Accordingly, an evaluation protocol must identify the target model and version, task, adversary access, input modality, dataset, metric, decision rule, and residual risk.

2.2 Deployment and application context

Safety is treated as a system property because model behaviour is affected by retrieval sources, external tools, quantisation, fine-tuning, and multi-agent interaction. For example, indirect injection can propagate through documents and web-agent context [59, 108], while adversarial behaviour can spread across interacting engineering agents [89]. These deployment surfaces extend the threat boundary beyond the user prompt and require controls at the input, model, orchestration, output, and operational layers.

Computational-efficiency and task-reliability research provides necessary deployment context, although it is not treated as direct evidence of adversarial safety. Transformer training and inference optimisations reduce memory, latency, and hardware costs [6, 20, 88, 102], while resource-aware architectures can preserve representational capacity under tighter computational constraints [36]. Studies of visual question answering, numerical reasoning, and time-aware recommendation further show that reliability depends on prompt formulation, reasoning structure, and temporal context [34, 42, 44]. Collectively, these studies establish two practical requirements for trustworthy deployment: safeguards must remain feasible within the available resource budget, and safety must be re-evaluated after architectural, optimisation, or task-context changes rather than inferred from average task performance.

Application context determines the consequence of failure. Healthcare studies associate hallucination or injected visual instructions with patient-safety and clinical-integrity risks [97, 105]; cybersecurity and software applications emphasise code integrity, tool misuse, and malicious-content detection [78, 92]; finance and business applications emphasise decision integrity and compliance [33]; and intelligent infrastructure adds availability and physical-system consequences [25, 89]. Multilingual deployments also require language-specific testing because semantic obfuscation and injection can evade safeguards developed primarily for English [81, 101].

The broader application literature also clarifies why safety criteria must be domain- and stakeholder-specific. Automated health monitoring and radiology-report generation raise requirements for clinical validity, privacy, and faithful communication [1, 61]; smart-contract analysis requires security-aware validation under imbalance and adversarial code patterns [80]; and sentiment analysis for Indonesian social-media data demonstrates the need to test language, culture, and distribution shift rather than assume uniform multilingual performance [63]. User-practice and political-economy studies additionally show that deployment risk is shaped by how people negotiate model behaviour and by institutional concentration, transparency, and accountability [56, 58]. Finally, the limitations of conventional vulnerability-scoring systems for LLM attacks reinforce the need to assess generative harms, exploitability, operational consequence, and residual risk together [9]. These works are therefore used to define deployment constraints and consequence-sensitive evaluation, while the security findings themselves are synthesised in Sect. 4.

2.3 Evaluation and governance context

Average-case accuracy is not a security guarantee. We recommend that safety evaluations measure attack success rate (ASR), harmful compliance, detection performance, over-refusal, and task utility under comparable threat conditions. Here, harmful compliance concerns model responses that fulfil harmful or policy-violating requests under the stated evaluation criteria; it is distinct from regulatory compliance. Recent defences illustrate complementary layers: query-and-response detection [90], structured injection-resistant pipelines [5], comparative prompt-injection detectors [22], and decontextualised output safeguards [76]. We also recommend connecting technical controls to audit logging, incident response, versioned re-evaluation, and human oversight. These concise definitions and assumptions form the basis of the taxonomy and evidence synthesis introduced in Sect. 4.

The assurance gates, validation milestones, reporting recommendations, and risk-priority rubric proposed in this

review are engineering recommendations derived from the evidence synthesis. An assurance gate denotes a proposed checkpoint for reviewing evidence and recording a deployment decision. These recommendations are not presented as statutory requirements, and adopting them does not establish legal compliance. The NIST AI RMF and its Generative AI Profile provide voluntary guidance [64, 65]. Any applicable legal obligations arise independently from the relevant legislation and depend on its scope, the actor's role, the system or model concerned, and the applicable provisions and dates [24].

3 Methodology

The review synthesises peer-reviewed evidence on foundation-model vulnerabilities, adversarial evaluation, mitigations, evaluation practice, and research gaps through the five research questions in Table 2. The protocol is reported in four stages: review design, search and selection, quality assessment and data extraction, and synthesis.

3.1 Review design and PRISMA-aligned reporting

Kitchenham's systematic-review guidance [40] was selected because it provides domain-appropriate procedures for research-question formulation, protocol design, study-quality assessment, data extraction, and evidence synthesis in software engineering and information-security research. PRISMA 2020 was used as a complementary reporting framework [68]: Kitchenham guided how the review was conducted, whereas PRISMA guided transparent reporting of identification, deduplication, screening, eligibility, quality exclusion, and final inclusion.

Figure 1 summarises the protocol. A pilot tested keyword relevance, eligibility decisions, and extraction fields before the final search. The search combined database retrieval, backward snowballing, and targeted venue screening; multiple authors checked ambiguous screening and coding decisions. Table 3 shows the database-specific queries for reproducibility.

3.2 Search and study selection

Searches covered January 2021–July 2026 and paired the model terms "large language model", "foundation model", and "Transformer" with the safety and adversarial terms recorded in Table 3: "robustness", "adversarial", "prompt injection", "jailbreak", "red team", and "hallucination". Queries were adapted to ACM Digital Library, IEEE Xplore, ScienceDirect, Scopus, and Google Scholar. Backward snowballing and targeted screening of EMNLP and NeurIPS supplemented the database search.

This review did not document forward snowballing or separate screening logs for USENIX Security, IEEE Symposium on Security and Privacy, and ACM CCS. These are therefore treated as search limitations rather than retrospectively claimed procedures. A future corpus update should execute and report those expansion streams with exact dates, strings, retrieved records, duplicates, exclusion reasons, and quality-assessed additions.

The search strings in Table 3 did not explicitly include dedicated terms for data poisoning, backdoors, model stealing, or model extraction. Although broader adversarial-search terms could retrieve relevant studies, the absence of these targeted terms, documented forward snowballing, and separate screening logs for the major security venues limits coverage of these research areas. The retained corpus should therefore not be treated as a comprehensive or fully representative sample of research on these attack surfaces.

Figure 2 reports the complete selection flow. The database search returned 382 records and 267 remained after deduplication; title-and-abstract screening retained 174 for full-text assessment, of which 63 met the main-search eligibility criteria. The pilot contributed 15 eligible studies and the secondary search contributed 23, giving 101 studies for quality assessment. Seven studies scoring below the quality threshold were excluded, leaving 94 for synthesis. Eligibility was assessed against Table 4, and exclusion reasons and source streams were retained for traceability.

Titles and abstracts were used only for retrieval and preliminary screening; inclusion required an accessible peer-reviewed full paper. Extended abstracts, standalone abstracts, editorials, and preprints without a peer-reviewed version were excluded. When a preprint and a version of record described the same study, only the peer-reviewed version was retained. Public standards, legislation, and developer safety frameworks cited later are explicitly identified as contextual sources: they inform terminology, governance, and industrial-practice comparison but are not counted in the 94-study corpus or its quantitative figures.

The first three authors checked potentially eligible records and resolved disagreements through discussion, following investigator triangulation [8]. The secondary-search stream returned 93 candidates; 57 remained after preliminary filtering, and 23 met the eligibility criteria.

3.3 Quality assessment and data extraction

The 101 eligible studies were assessed against five adapted criteria [55]: clarity and relevance of the aim, appropriateness of the research design, clarity of findings, reporting of limitations, and evidential support for safety or mitigation claims. Each criterion received 1, 0.5, or 0 points. Studies scoring at least 3.0 were classified as *Good*, those scoring from 2.0 to

Table 2 RQs on safety and robustness of LLMs and foundation models

No.	Research question
RQ1	What are the primary safety and robustness vulnerabilities (e.g., prompt injection, jailbreaking, harmful-content generation, hallucinations) identified in LLMs and other foundation models?
RQ2	What red-teaming methodologies, techniques, and tools are proposed or utilised in literature to systematically uncover these vulnerabilities?
RQ3	What technical and procedural solutions or mitigation strategies (e.g., adversarial training, instruction tuning, input/output filtering) are presented to enhance the safety and robustness of these models?
RQ4	How is the effectiveness of safety measures and red-teaming efforts evaluated and measured in the selected studies?
RQ5	What are the most significant research gaps and future directions identified for ensuring the safety of LLMs and other understudied AI systems?

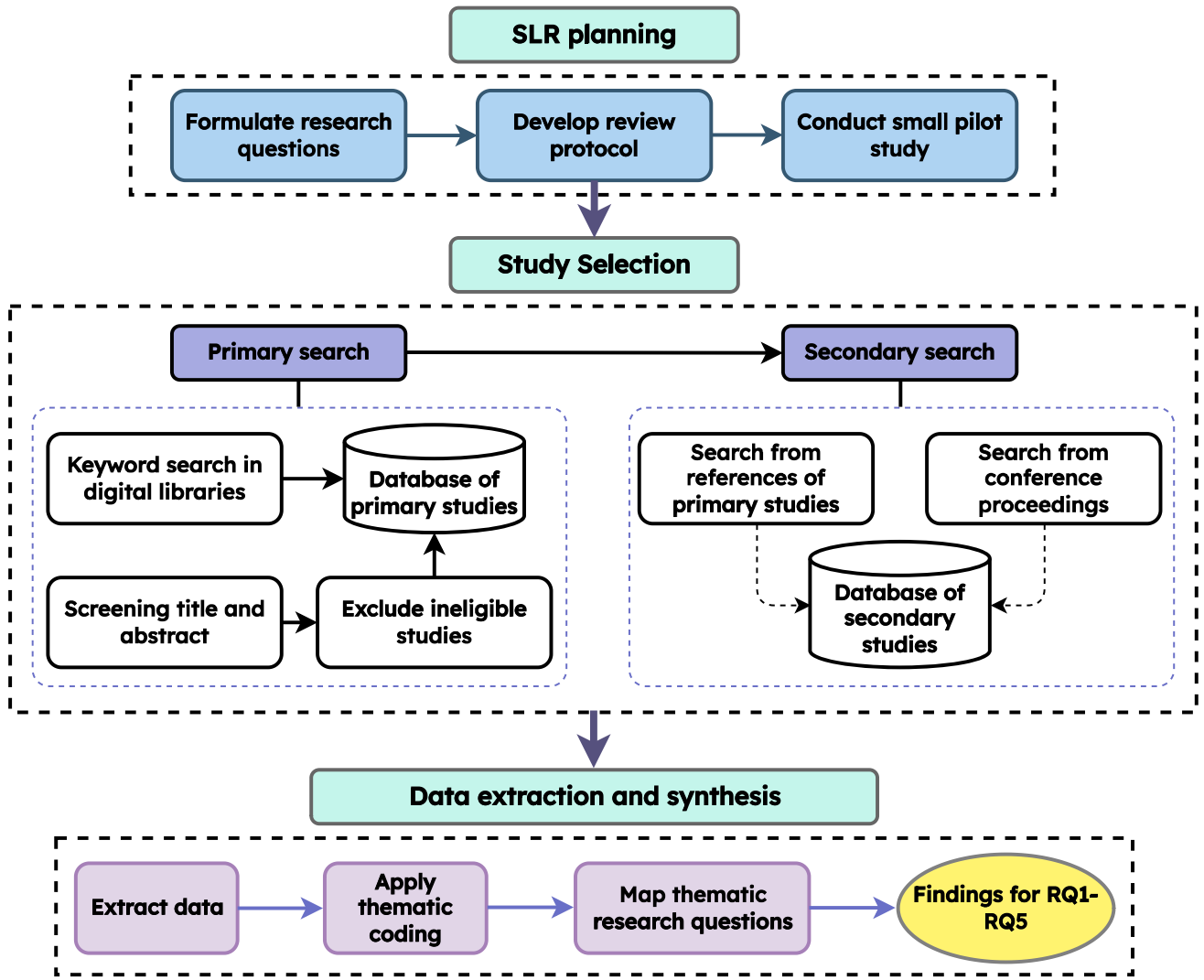


Fig. 1 Review protocol: preparation, conduct, quality assessment, extraction, synthesis, and reporting

Table 3 Search protocol for digital libraries

Digital library	Search string	Search within	Time frame
ACM Digital Library	([Abstract:"large language model*" OR Abstract:"foundation model*" OR Abstract: "transformer*"]) AND ([Abstract: "robustness"] OR [Abstract: "prompt injection"] OR [Abstract: "jailbreak*"] OR [Abstract: "adversarial"] OR [Abstract: "red team*"] OR [Abstract: "hallucination"])	Research article (Abstract)	January 2021–July 2026
IEEE Xplore	("Abstract": "large language model*" OR "Abstract": "foundation model*" OR "Abstract": "transformer*") AND ("Abstract": "robustness" OR "Abstract": "adversarial" OR "Abstract": "prompt injection" OR "Abstract": "jailbreak" OR "Abstract": "red team*" OR "Abstract": "hallucination")	Conference, Journal (Abstract)	January 2021–July 2026
ScienceDirect	("large language model*" OR "foundation model*" OR "transformer*") AND (robustness OR "adversarial" OR "prompt injection" OR "jailbreak*" OR "red team*" OR "hallucination")	Research articles (Title, Abstract, Keywords)	January 2021–July 2026
Scopus	TITLE-ABS-KEY(("large language model*" OR "foundation model*" OR "transformer*") AND ("robustness" OR adversarial OR "prompt injection" OR "jailbreak*" OR "red team*" OR "hallucination"))	Conference paper, Article (Title, Abstract, Keywords)	January 2021–July 2026
Google Scholar	("large language model*" OR "foundation model*" OR "transformer model") ("robustness" OR "prompt injection" OR "jailbreak" OR "red team*" OR "hallucination")	All (Title, Abstract, Keywords)	January 2021–July 2026

Table 4 Inclusion and exclusion criteria

Category	Criteria description
Inclusion criteria	Peer-reviewed conference papers and journal articles
	Studies published between January 2021–July 2026, capturing the contemporary LLM/foundation-model era
	Papers written in English
	Studies primarily focused on safety, robustness, alignment, or red-teaming/adversarial evaluation of LLMs or foundation AI systems
	Papers presenting empirical, methodological, or framework-based approaches related to risk identification, vulnerability testing, or mitigation strategies
	Research evaluating or discussing technical, procedural, or human-in-the-loop safety mechanisms for AI models
Exclusion criteria	Non-peer-reviewed materials (e.g., blog posts, theses, dissertations, preprints, editorials, or patents)
	Papers not written in English
	Studies where safety or robustness is only a minor or incidental aspect (e.g., performance optimisation or dataset construction papers)
	Duplicate studies, such as conference versions of extended journal papers
	Incomplete papers (e.g., unavailable full text)
	Papers focusing solely on bias/fairness, ethics, or sociotechnical aspects without direct relation to technical safety or robustness mechanisms

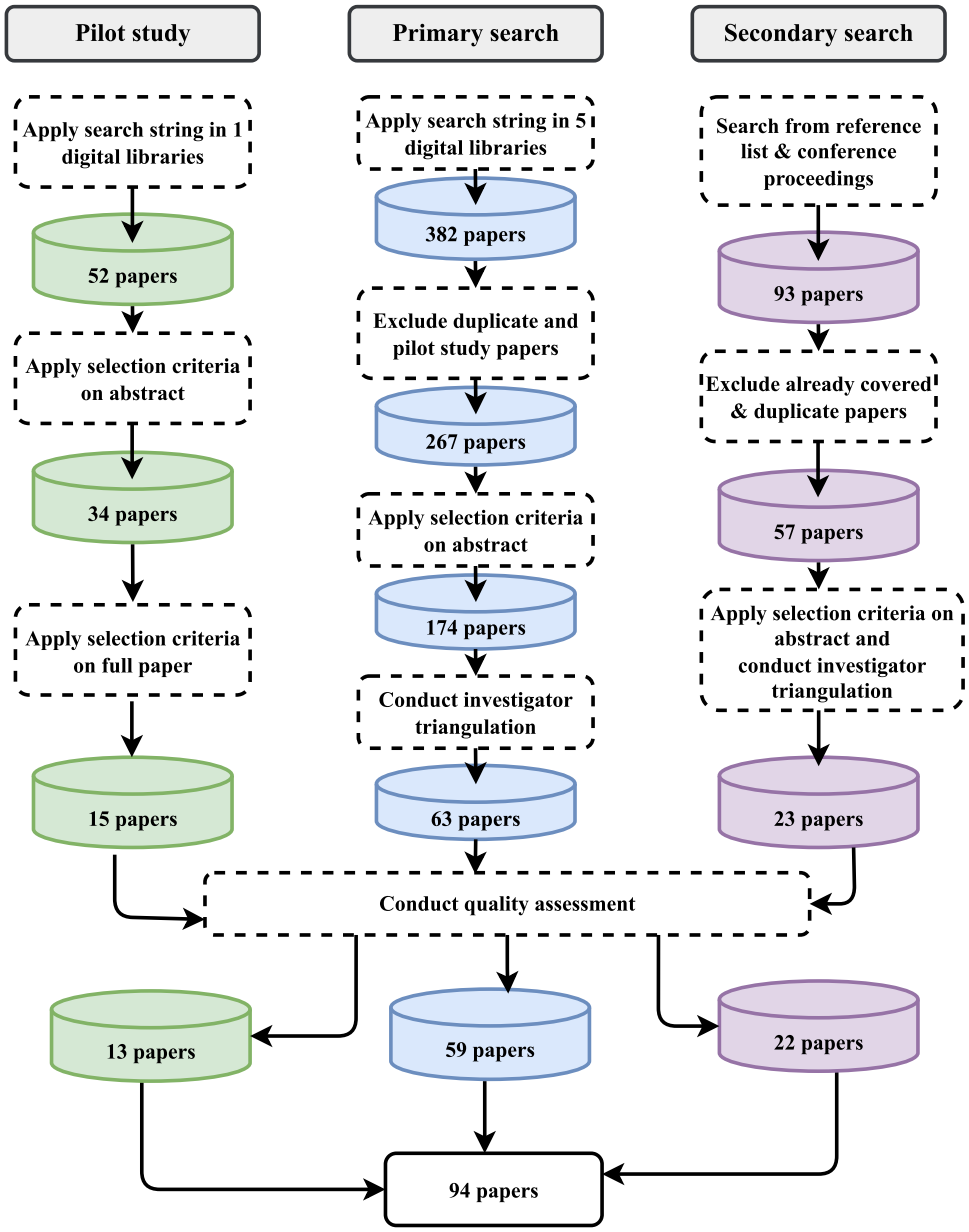


Fig. 2 PRISMA-aligned flow of identification, deduplication, screening, eligibility assessment, quality assessment, and final inclusion

below 3.0 as *Fair*, and those below 2.0 as *Poor*. Seven Poor studies were excluded, producing the final 94-study corpus.

Quality scores controlled how evidence was used after selection; they were not pooled with outcome measures. Good studies could anchor a synthesis claim, whereas Fair studies were used as corroborative or contextual evidence and were not treated as the sole basis for a prevalence or effectiveness conclusion. Poor studies contributed neither counts nor narrative conclusions because they were excluded before synthesis. Wherever the evidence permitted, a general claim was cross-checked against at least two independent included studies. Findings supported by only one study are identified as representative or illustrative and are not generalised to the corpus. Table 5 records these rules explicitly.

The structured extraction form recorded bibliographic data, model family and modality, application domain, benchmark, vulnerability and attack type, adversary access, red-teaming method, defence, evaluation metric and modality, reported result, limitation, and future-work statement. Missing fields were marked as not reported rather than inferred. Table 6 maps the retained fields to the RQs.

The first author performed the initial extraction; the second and third authors cross-checked completeness and classification. Calibration discussions resolved ambiguous multi-label cases, and the consolidated sheet generated the descriptive statistics, tables, and synthesis in Sect. 4.

Table 5 Quality-tier rules used to control evidential weight in the synthesis

Tier	QA score	Status in the review	Permitted evidential use
Good	≥ 3.0	Included in the 94-study synthesis	May anchor a mechanism, comparison, or cross-study conclusion; quantitative values remain conditional on the reported model, dataset, threat model, and evaluation design
Fair	2.0 to < 3.0	Included with qualified weight	May corroborate or contextualise a conclusion, but is not used alone to infer prevalence, superiority, or deployment readiness
Poor	< 2.0	Excluded before synthesis ($n = 7$)	Contributes no frequency count, metric range, effectiveness claim, or research conclusion
Single-study evidence	Good or Fair	Retained only as study-specific evidence	Labelled illustrative or representative; its numerical result is not pooled or presented as a field-wide estimate

Table 6 Data-extraction fields used for the systematic review

Field group	Extracted variables	Primary RQ link
Bibliographic and corpus data	Study identifier, title, authors, year, venue, publication type, publisher, author affiliation, country, citation count, and search source	Corpus overview
Model, task, and domain	Model family, modality, task, application domain, dataset, benchmark, and deployment context	RQ1, RQ4
Vulnerability and attack evidence	Reported safety issue, attack vector, adversary access, input channel, affected security property, and domain-specific failure mode	RQ1, RQ2
Red-teaming and evaluation design	Red-teaming method, attack-generation procedure, benchmark, evaluator type, metric, sample size, and automated or human-evaluation setting	RQ2, RQ4
Defence and mitigation evidence	Prompt-level, model-level, retrieval-level, filtering, verification, fine-tuning, monitoring, or human-in-the-loop defence mechanism; pre- and post-defence results where available	RQ3, RQ4
Limitations and research gaps	Reported limitations, residual risks, reproducibility concerns, deployment constraints, and future-work recommendations	RQ5

3.4 Data synthesis and analysis

Open codes were grouped through axial coding into higher-level vulnerability, attack, red-teaming, defence, evaluation, and research-gap categories. Coding was multi-label because one study could address several attacks, defences, domains, or gaps. Descriptive counts summarised the coded distributions, while narrative synthesis compared mechanisms, contexts, and residual risks. A meta-analysis was not performed because models, tasks, datasets, adversary access, attack strength, metrics, and judgement procedures differed substantially across studies.

To prevent category overlap from being mistaken for inconsistent coding, the codebook separates orthogonal dimensions. In particular, *multimodal* denotes the input or fusion channel through which an attack is delivered, whereas the *application domain* records the context and consequence of the failure. A visual injection in surgical decision support is therefore coded simultaneously as a multimodal attack and a healthcare exposure, not placed ambiguously into one category. The same rule separates an attack mechanism from its lifecycle surface, attacker capability, affected security property, and application domain.

Operationally, a failure without purposeful manipulation was coded as a vulnerability, whereas an attack code required an adversarial action, payload, optimisation, or trigger. A multimodal code required a manipulated non-text or cross-modal signal; an agent/workflow code required exploitation of retrieval, memory, tools, authorisation, inter-agent communication, or orchestration. Data poisoning denotes malicious modification of training, fine-tuning, or retrieval data; a backdoor requires trigger-conditioned behaviour introduced through data, adaptation, or compromised weights; and model stealing denotes unauthorised extraction or functional imitation of model behaviour or assets [86]. The unit of counting was the retained study: a study was counted at most

Table 7 Orthogonal coding dimensions and decision rules

Dimension	Coding question	Decision rule and examples	Analytical role
Lifecycle attack surface	Where is adversarial influence introduced or amplified?	Training/data supply chain; model or configuration; instruction interface; retrieval, memory, or tools; multimodal fusion; agent or workflow orchestration	Defines the hierarchical attack-surface taxonomy for RQ1
Attack mechanism	How is the security objective pursued?	Jailbreak, direct or indirect injection, evasion, poisoning or backdoor, model extraction or stealing, adversarial fine-tuning, or unauthorized tool invocation. Mechanisms may occur at several surfaces	Prevents conflating a mechanism with its deployment location
Modality	Through which channel is the malicious signal conveyed?	Text, code, image, audio, video, or mixed modality. "Multimodal" is coded here, independent of application domain	Supports modality-specific attack and defence comparisons
Application domain	Where does the resulting failure have consequences?	Healthcare, cybersecurity/software, business/finance, intelligent infrastructure, multilingual/social, or general/agentic use	Connects technical findings to domain-specific harm and assurance needs
Adversary capability	What access, knowledge, and budget are assumed?	Black-, grey-, or white-box access; direct or indirect control; query/optimisation budget; transfer and adaptivity	Conditions ASR and robustness claims and supports reproducibility
Security consequence	Which property and harm are affected?	Confidentiality, integrity, availability, privacy, safety, accountability, or operational reliability; reversible output error is distinguished from persistent, physical, or large-scale harm	Supports context-dependent risk prioritisation rather than unqualified severity labels

once within a category regardless of its number of models or experiments, but could receive several distinct codes when each operational rule was satisfied.

For the lifecycle attack-surface axis, 52 of the 94 retained studies met the adversarial coding criteria; the remaining 42 were uncoded on this axis. Category-specific counts include each study at most once per category, while allowing assignments to several categories. Percentages are reported against both the complete corpus ($n = 94$) and the coded subset ($n = 52$), with the denominator identified explicitly. This axis is distinct from the broader vulnerability-category coding summarised for the full corpus in Figs. 7 and 8.

The final codebook was refined by repeated comparison with RQ1–RQ5 and used to generate the results in Sect. 4. Table 8 makes the path from extracted evidence to coding category and synthesis output explicit.

3.5 Study characteristics and corpus overview

This section summarises selected studies, providing context for the state of the art and trends in the review corpus. The analysis investigates the temporal spread of papers, their distribution across leading digital libraries, and the citation impact of all included studies. Such descriptive views inform the forthcoming results by highlighting trends in research activity, the diversity of publication outlets, and the literature's stage of evolution. Together, they provide a basis for understanding the vulnerability, attack, countermeasure, assessment, and evaluation practices presented in the following sections.

Figure 3 shows the contribution of each search source to the final corpus: ScienceDirect supplied 24 studies (25.5%), IEEE Xplore 21 (22.3%), ACM Digital Library 17 (18.1%), Google Scholar 17 (18.1%), and Scopus 15 (16.0%). These values represent the retained source recorded after deduplication, so each of the 94 studies is counted once.

Figure 4 separates the corpus into five citation ranges: 22 studies had no citations (23.4%), 36 had 1–10 (38.3%), 23 had 11–50 (24.5%), four had 51–100 (4.3%), and nine had more than 100 (9.6%) at the extraction date. The concentration in the lower ranges is consistent with the corpus's recency and should not be interpreted as a quality score.

Table 8 Mapping between research questions, extracted evidence, coding categories, and synthesis outputs

RQ	Extracted evidence	Main coding categories	Synthesis output
RQ1	Reported safety and robustness failures	Lifecycle surface, attack mechanism, modality, adversary capability, affected security property, and application domain coded as separate dimensions	Hierarchical attack-surface taxonomy, vulnerability distribution, and domain-consequence mapping
RQ2	Red-teaming procedures and adversarial evaluation methods	Manual red-teaming, automated prompt generation, benchmark-based attacks, black-box attacks, white-box attacks, multimodal attacks, transfer attacks	Red-teaming-method distribution and attack-method taxonomy
RQ3	Technical and procedural mitigation strategies	Prompt-level defences, filtering, adversarial training, robust fine-tuning, RAG grounding, cross-model verification, gatekeeper models, human oversight	Defence taxonomy and attack–mitigation mapping
RQ4	Evaluation metrics, protocols, and judgement procedures	Attack success rate, robust accuracy, hallucination rate, F1/AUC/accuracy, precision/recall, expert evaluation, automated evaluation, hybrid evaluation	Metric synthesis and evaluation-modality comparison
RQ5	Reported limitations and future-work directions	Benchmark fragmentation, multilingual gaps, low-resource evaluation, adaptive adversaries, long-term deployment, governance, reproducibility	Research-gap synthesis and future-direction mapping

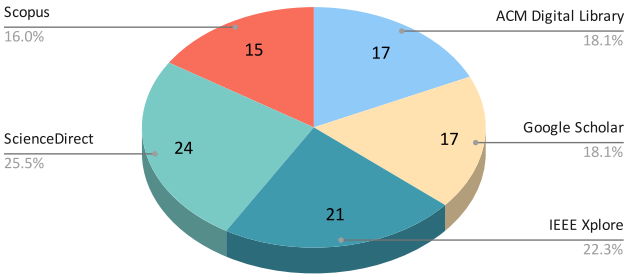


Fig. 3 Distribution of the 94 selected studies across digital libraries

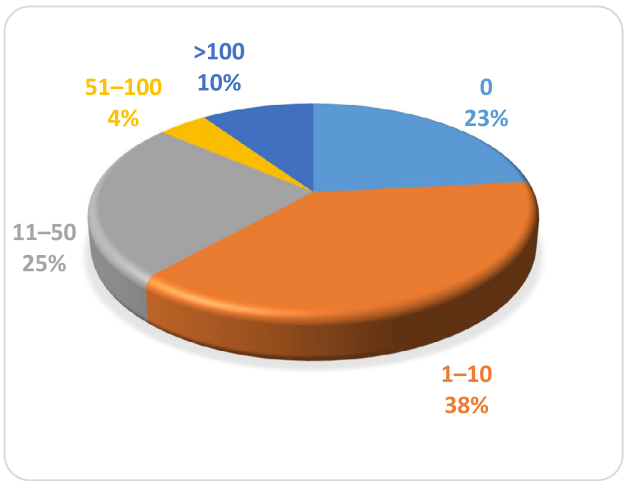


Fig. 4 Citation distribution of the 94 reviewed studies by citation range

Figure 5 reports 3 studies from 2021, 5 from 2022, 10 from 2023, 16 from 2024, 38 from 2025, and 22 from January–July 2026. Thus, 60 of the 94 studies (63.8%) were published from 2025 onward; the 2026 value represents only the seven-month search window and is not directly comparable with a complete calendar year.

4 Results and discussion

This section reports the findings of the SLR against RQ1–RQ5. The synthesis is organised around the main evidence categories extracted from the 94 quality-screened studies: reported vulnerabilities and attack surfaces; red-teaming and adversarial evaluation methods; defence and mitigation mechanisms; evaluation metrics; high-stakes validation; and unresolved research gaps. The results are interpreted from an information-security perspective, with emphasis on adversary behaviour, deployment context, defence-in-depth, reproducibility, and residual risk.

Figure 6 presents the scope of the review as an iterative security lifecycle rather than a collection of independent attacks and defences. Autonomous red-teaming broadens adversarial testing [30, 92]; layered and self-evolving defences connect discovered attacks to revised controls [73, 90]; and retrieval- and agent-oriented studies show why monitoring and re-testing must cover the complete deployed system [59, 108].

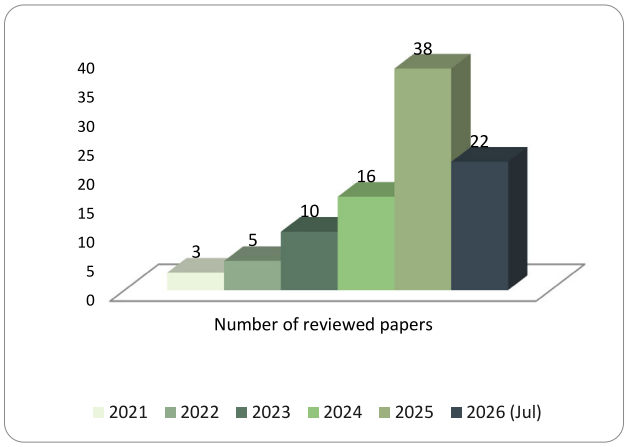


Fig. 5 Year-wise distribution of the 94 reviewed studies (2026 covers January–July only)

4.1 Synthesis approach and RQ-level distributions

The reviewed studies show that LLM and foundation-model safety is a multidimensional security problem rather than a collection of isolated failure cases. We therefore removed the unconditioned cross-study metric ranges. Table 9 instead provides an author-proposed reporting and assessment framework: an ASR, hallucination rate, robust-accuracy value, or detector score is interpretable only together with the target system, threat model, data, attack intensity, benign utility, operational cost, judgement procedure, and evidence tier. Values from incompatible experimental conditions are not averaged or ranked.

For RQ1, Fig. 7 quantifies vulnerability breadth: 46 studies report one coded vulnerability category, 34 report two, nine report three, two report four, and three report five. Figure 8 therefore separates 46 single-vulnerability studies (48.9%) from 48 multi-vulnerability studies (51.1%). These figures describe vulnerability-category breadth across all 94 studies. They do not report the frequency of individual lifecycle attack surfaces. The separate lifecycle axis in Table 10 has 52 coded studies, and its category-specific counts and percentages are reported in Sect. 4.2. Together, the results show that prompt injection, jailbreaks, hallucination, adversarial manipulation, multimodal attacks, and deployment-specific misuse frequently require joint analysis.

For RQ2, Fig. 9 shows that 13 studies use one red-teaming method, 22 use two, 28 use three, 13 use four, nine use five, five use six, and four use seven. Figure 10 consequently identifies 13 single-method studies (13.8%) and 81 multi-method studies (86.2%). This concentration indicates a move from isolated prompt testing toward combinations of manual prompting, automated generation, benchmark-based testing, transfer attacks, and multimodal or agentic attacks [30, 32, 92].

For RQ3, Fig. 11 shows that 15 studies report no mitigation strategy, 41 report one, 26 report two, seven report three, and five report four. Figure 12 therefore groups the corpus into 15 no-strategy studies (16.0%), 41 single-strategy studies (43.6%), and 38 multi-strategy studies (40.4%). Compared with the 81 studies using multiple red-teaming methods, this distribution reveals a gap between breadth of attack discovery and breadth of defence evaluation. Recent layered controls

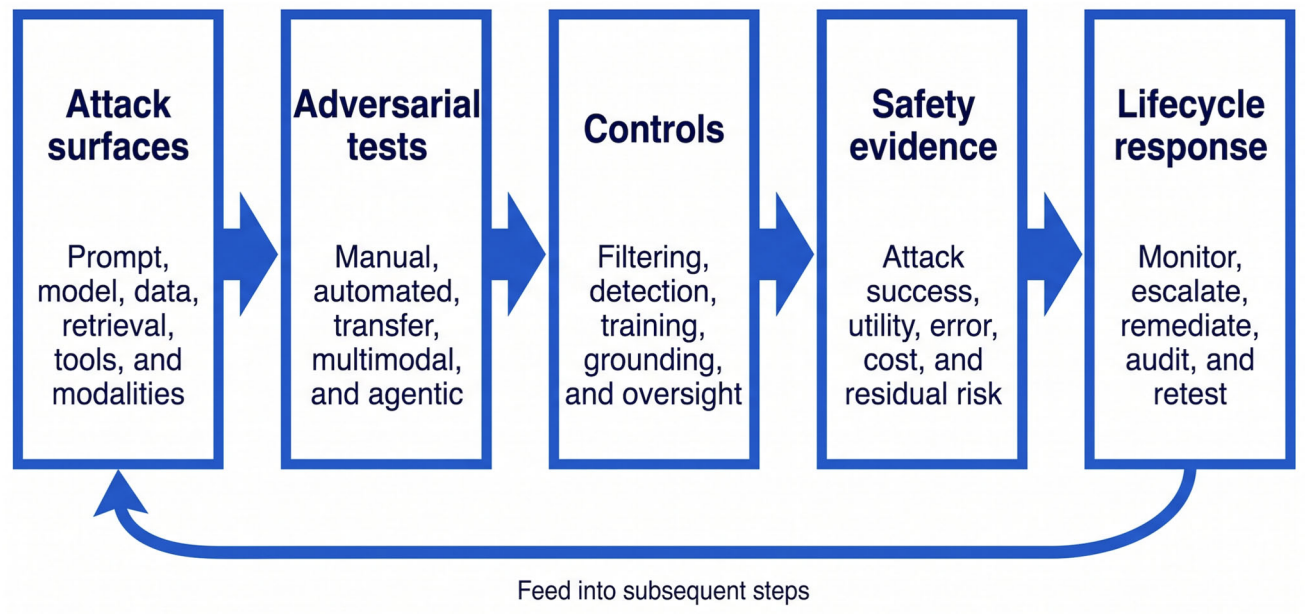


Fig. 6 End-to-end synthesis of the security lifecycle covered by the review. Post-deployment evidence feeds subsequent threat models, adversarial tests, and control updates

Table 9 Unified cross-domain safety-assessment and reporting framework

Assessment component	Recommended experimental reporting fields	Core measures	Interpretation rule
Target system	Provider, model and policy version, parameter scale when reported, precision or quantisation, system prompt, retrieval/tool configuration, and release date	Configuration manifest and version identifier	Do not combine results across materially different model or deployment configurations
Threat model and attack intensity	Attacker goal, black-/grey-/white-box access, controlled channel, adaptivity, number of turns, query or optimisation budget, perturbation bound, and transfer setting	ASR, harmful-compliance rate, bypass rate, data-attack success, and time or queries to failure	Compare security effectiveness only under equivalent attacker capability and budget
Data and task	Dataset and version, sample size, task, domain, modality, language, sampling protocol, benign/adversarial split, and baseline	Robust accuracy, hallucination or factual-error rate, calibration, task accuracy, F1, AUC, precision, and recall	Disaggregate by domain, language, modality, and attack family; report uncertainty where the design permits
Defence effectiveness and residual risk	Defence layer, intervention threshold, pre-/post-defence protocol, adaptive re-test, transfer test, and uncaught failure definition	Absolute and relative ASR reduction, false-negative rate, post-adaptation ASR, and residual-risk statement	A lower attack rate is not a complete result unless the remaining failure modes and adaptive tests are reported
Benign utility and side effects	Clean-task baseline, refusal policy, user population, domain harm threshold, and intervention or escalation rule	Utility retained, over-refusal, false-positive rate, calibration, expert agreement, and subgroup effects	Security improvement must be interpreted with utility loss and unequal error or refusal burdens
Operational feasibility	Hardware or service setting, number of model calls, token and memory use, throughput requirement, and human-review workload	Latency, compute, memory, energy, API/query cost, and analyst time	Report marginal overhead against the undefended system and identify the deployment scale at which the control remains feasible
Evidence and decision	Evaluator type, annotation protocol, confidence interval or sensitivity analysis, study quality tier, independent corroboration, and decision owner	Human/automated agreement, uncertainty, replication, incident evidence, and risk-acceptance decision	Good and independently corroborated evidence receives greatest weight; single-study results remain illustrative

Note: The reporting fields are recommended by the authors to support comparability and reproducibility. This framework is not presented as a regulatory conformity-assessment or certification scheme

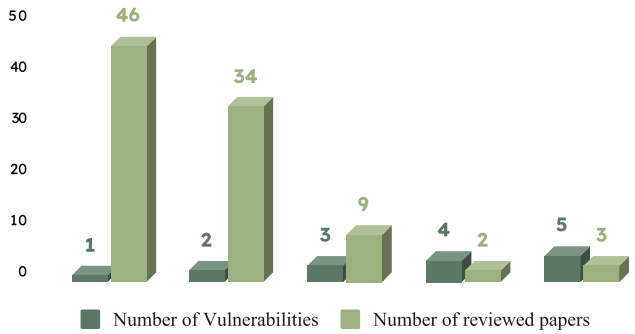


Fig. 7 Number of vulnerability categories per study (one: 46; two: 34; three: 9; four: 2; more than four: 3)

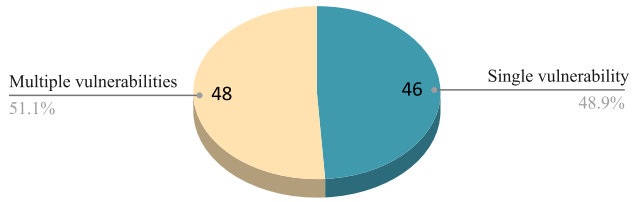


Fig. 8 Single- versus multi-vulnerability studies in the 94-study corpus (46 versus 48)

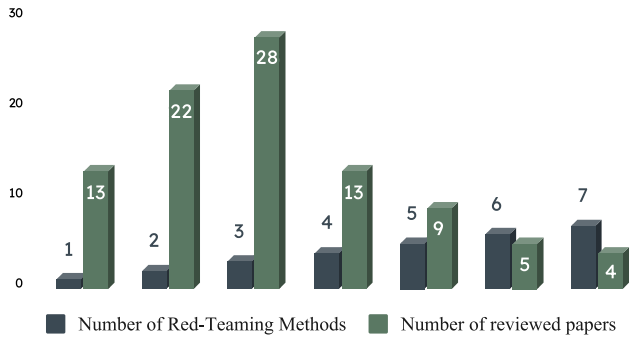


Fig. 9 Number of red-teaming methods reported per study in the 94-study corpus

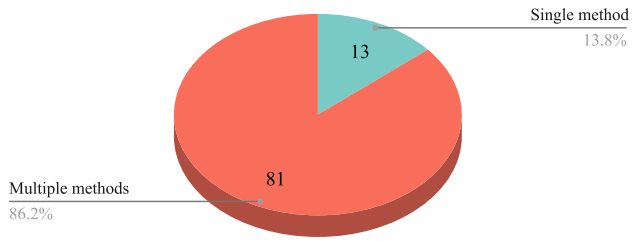


Fig. 10 Single- versus multi-method red-teaming studies (13 versus 81)

nevertheless show movement toward defence-in-depth [5, 73, 76, 90].

For RQ4, Figs. 13 and 14 separate reported metric families from evaluation modalities. Of the 94 studies with a classifiable primary metric, 55 used qualitative/case analysis, 16 used accuracy/F1/AUC, 11 used human/expert evaluation as

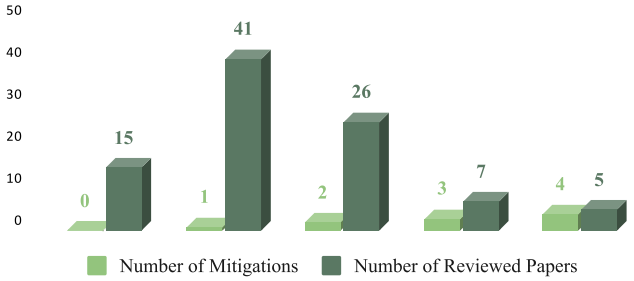


Fig. 11 Number of mitigation strategies reported per study in the 94-study corpus

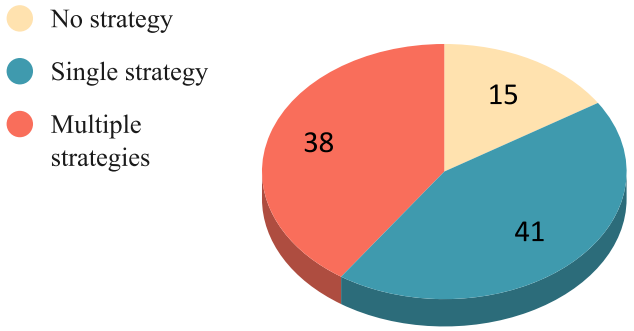


Fig. 12 No, single, and multiple mitigation strategies (15, 41, and 38 studies)

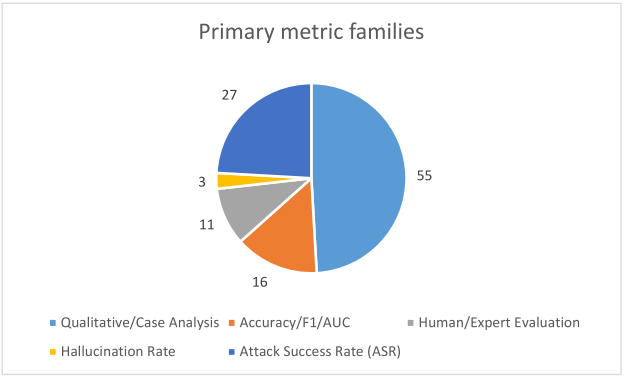


Fig. 13 Primary metric families ($n = 112$) from 94 papers: qualitative/case analysis 55; accuracy/F1/AUC 16; human/expert 11; hallucination rate 3; ASR 27

the metric family, three used hallucination rate, and 27 used ASR. Of the 69 studies with a classifiable evaluation modality, 61 were automatic/algorithmic only (88.4%), five were hybrid (7.2%), and three were human/expert only (4.3%); the remaining 25 did not report a classifiable evaluator type. This imbalance is especially consequential in clinical, cybersecurity, and infrastructure settings, where an automatically acceptable output may still be unsafe or operationally unusable.

For RQ5, Fig. 15 makes the qualitative synthesis explicit by mapping each recurring evidence gap to its security implication and a corresponding future-work programme. The five

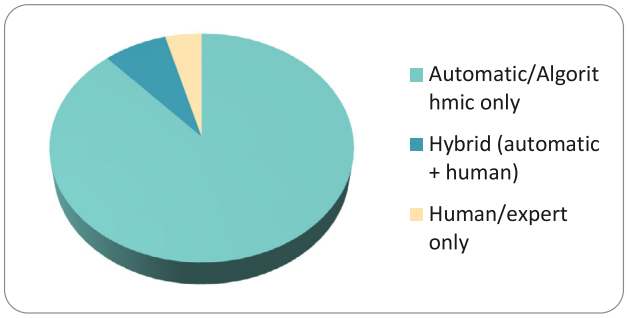


Fig. 14 Evaluation modalities ($n = 69$): automatic/algorithmic only 61; hybrid 5; human/expert only 3. Twenty-five studies did not report a classifiable evaluator type

themes are benchmark comparability, coverage across languages and modalities, adaptive and longitudinal evaluation, cross-layer defence, and operational governance. Recent multilingual, autonomous, indirect-injection, web-agent, and clinical multimodal studies sharpen these gaps but do not eliminate them [30, 59, 81, 101, 105, 108].

4.2 Cross-layer attack and application taxonomy

Rather than treating the reviewed attacks as an unstructured list, we constructed the hierarchical, multi-label taxonomy in Table 10 by combining the corpus coding procedure in Sect. 3 with the NIST adversarial-machine-learning terminology [86]. Its first axis identifies the lifecycle boundary at which adversarial control is introduced or amplified; its second axis identifies the application in which the resulting failure has consequences. The structure distinguishes training-data poisoning or a model-asset attack from an instruction attack, compromised external context, multimodal manipulation, or agentic propagation, even when each ultimately produces an unsafe response.

The orthogonal application axis comprises (i) healthcare and clinical decision support, (ii) cybersecurity and software engineering, (iii) business, finance, and knowledge work, (iv) intelligent infrastructure and engineering, (v) multilingual and social-media services, and (vi) general-purpose and agentic systems [25, 33, 78, 101, 105, 108]. Because categories are non-exclusive, one study may receive several lifecycle, mechanism, modality, and application codes. This preserves compound attacks and shows that agentic systems do not replace prompt-level weaknesses; they can propagate them through retrieval, memory, tools, and interacting agents [30, 53, 92]. Standards and governance documents used to align terminology or interpret deployment obligations are contextual sources and are not included in the 94-study frequency counts.

Applying the lifecycle attack-surface coding to the 94 included studies produced 74 assignments across 52 studies (55.3%) that evaluate at least one adversarial attack surface, equivalent to a mean of 1.42 lifecycle categories per coded study. The remaining 42 studies (44.7%) address capability, efficiency, benchmarking, or domain-application questions

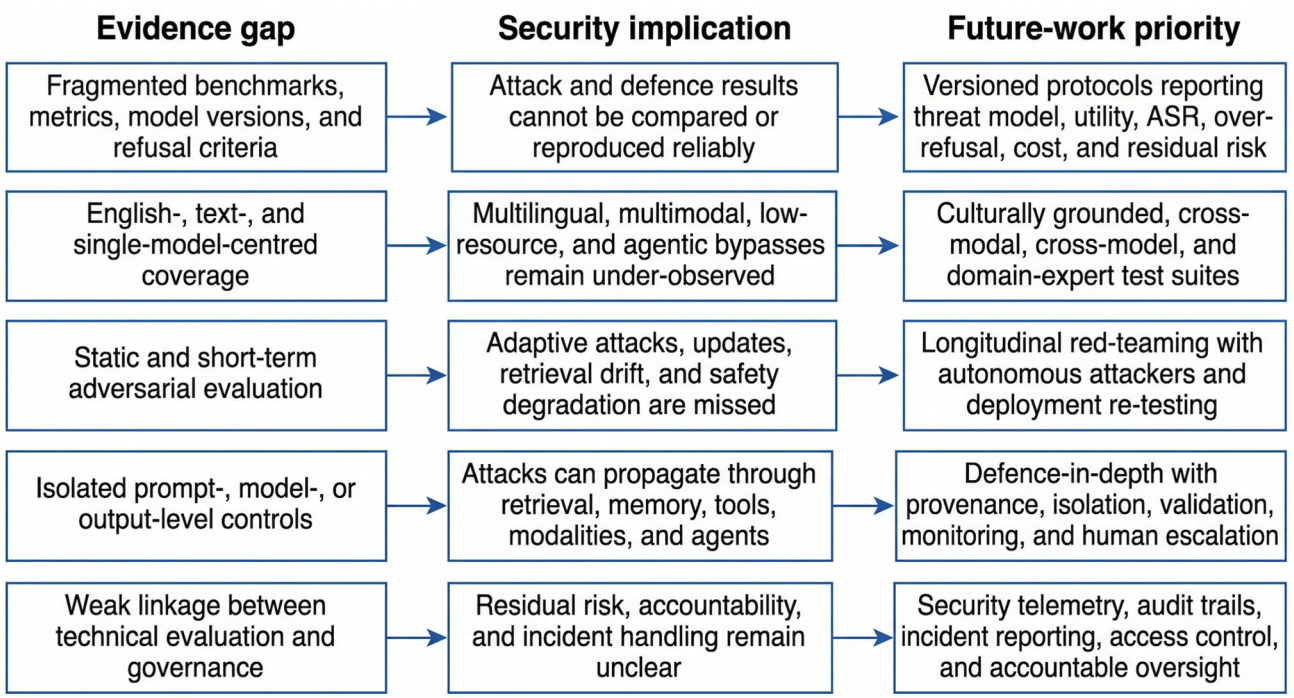


Fig. 15 Evidence-to-action roadmap linking the principal research gaps to security consequences and future work

Table 10 Hierarchical lifecycle taxonomy linking foundation-model attack surfaces to security boundaries and application exposures

Taxonomy level	Attack mechanisms and representative evidence	Compromised boundary	Application exposure and characteristic consequence
Level 1: Training and data supply chain	Training-data or retrieval poisoning, backdoor insertion, malicious adaptation data, and provenance compromise [74, 86]	Training corpus, adaptation data, retrieval index, and data provenance	All application domains: persistent or trigger-dependent manipulation that may survive ordinary clean-data evaluation
Level 2: Model assets and deployment configuration	Model extraction or stealing, parameter or embedding-space attacks, malicious fine-tuning, and safety degradation after quantisation or other deployment changes [13, 49, 86]	Weights, model API, alignment behaviour, and configuration-dependent safeguards	Open-weight, API, edge, cybersecurity, finance, and infrastructure deployments: confidentiality loss or post-deployment loss of robustness
Level 3: Instruction and interaction	Direct injection, jailbreak prompting, semantic obfuscation, reasoning-continuation attacks, and multilingual bypass [21, 28, 53, 101, 104]	Instruction hierarchy, policy compliance, and user–model interaction	General assistants, business and finance services, clinical chat, and multilingual services: unsafe compliance, misinformation, or decision manipulation
Level 4: External context, memory, and tools	Indirect injection through documents, webpages, retrieval sources, memory, tool output, or concealed instructions [59, 83, 108]	Context provenance, retrieval integrity, and separation of trusted and untrusted content	RAG applications, web agents, peer review, and enterprise or clinical knowledge services: corrupted evidence, leakage, or attacker-controlled output
Level 5: Multimodal fusion	Visual prompt injection, adversarial images, transferable visual perturbations, and cross-modal inconsistency [50, 105, 106]	Visual encoder, modality-fusion mechanism, and cross-modal consistency	Surgical and clinical decision support, social-media analysis, and industrial vision: health, reputational, or physical consequences
Level 6: Agent and workflow orchestration	Autonomous jailbreak agents, context-aware attack generation, multi-agent prompt fusion, and propagation through tools or interacting agents [30, 32, 92]	Agent authorisation, memory, tool-use policy, and inter-agent trust	Software and cybersecurity operations, web agents, engineering workflows, and enterprise automation: unauthorised actions, propagation, or loss of human control

Table 11 Illustrative author-proposed, context-conditioned risk-priority rubric for the attack-surface taxonomy

Factor	Score 1	Score 2	Score 3
Impact (I)	Reversible task error with limited downstream consequence	Material privacy, integrity, financial, or operational harm requiring intervention	Potential patient, physical, critical-infrastructure, large-scale, or persistent supply-chain harm
Exploitability (E)	White-box access, high expertise, or substantial optimisation is required	Limited internal knowledge, repeated interaction, or moderate resources are required	Black-box, low-cost, transferable, automated, or readily repeatable exploitation
Exposure (X)	Isolated/offline model with no sensitive data or action privilege	User-facing system with bounded data, retrieval, or tool access	Public, high-stakes, autonomous, multi-agent, or privileged tool-enabled deployment

Note: This rubric is an illustrative, author-proposed triage aid only. It is neither empirically validated nor standardised and does not determine regulatory risk classification. Its scoring criteria, equal weights, and priority thresholds require domain-specific adaptation and validation before operational use

without evaluating an adversarial attack surface that meets this axis's coding criteria. They therefore remain uncoded on this axis; this does not imply that the systems they examine are free from security vulnerabilities.

In the following distribution, the first percentage uses the complete corpus of 94 studies as its denominator, and the second uses the coded subset of 52. Level 1, training and data supply chain, includes 3 studies (3.2%; 5.8%), only one of which evaluates poisoning empirically. Level 2, model assets and deployment configuration, includes 10 studies (10.6%; 19.2%), concentrated on parameter- and embedding-space attacks, safety degradation after quantisation, and robustness loss introduced during adaptation; model extraction or distillation-based stealing appears in only two studies. Level 3, instruction and interaction, is the most extensively represented category, comprising 44 studies (46.8%; 84.6%) addressing direct injection, jailbreak prompting, semantic obfuscation, reasoning-continuation attacks, multilingual bypass, and corresponding defences. Level 4, external context, memory, and tools, includes 4 studies (4.3%; 7.7%); Level 5, multimodal fusion, includes 6 (6.4%; 11.5%); and Level 6, agent and workflow orchestration, includes 7 (7.4%; 13.5%).

Each study is counted at most once within a lifecycle category but may contribute to several categories. Consequently, the 74 assignments represent 142.3% of the 52-study coded subset and 78.7% of the complete 94-study corpus. These totals reflect overlapping assignments and the presence of studies uncoded on this axis. The distribution describes coverage within the retained corpus, rather than the prevalence or severity of attacks in deployed systems. Counts for the broader training/data and model-assets categories should not be interpreted as counts specifically for poisoning, backdoors, or model stealing.

Data poisoning, backdoors, and model stealing are retained in Table 10 for taxonomic completeness and alignment with established adversarial-machine-learning terminology. However, empirical evidence for these mechanisms within the 94-study corpus is thin, reflecting the search-strategy limitations documented in Sect. 3.2. The present corpus should therefore not be treated as fully representative of these threat categories. Their limited representation is a limitation of this review's coverage and does not demonstrate low prevalence, low severity, or a general absence of research.

Table 11 presents an illustrative, author-proposed triage aid for preliminary comparison of attack scenarios. By considering impact, exploitability, and exposure together, the rubric provides a structured way to identify threats that may warrant greater attention within a specified operational context. It supports exploratory risk screening and does not provide a definitive assessment of security risk. The additive score $R = I + E + X$, equal weighting scheme, and proposed thresholds of 3–4 (lower priority), 5–6 (moderate priority), and 7–9 (high priority) have not been empirically validated and do not constitute a standardised risk-assessment model. Because the relative importance of the three components may differ across healthcare, finance, critical infrastructure, and other domains, these weights and thresholds should not be adopted directly for real-world risk evaluation. Any practical application requires domain-specific adaptation, empirical validation, and alignment with the organisation's established risk-management process. Assessors should document the evidence supporting each component. The rubric is conceptually informed by the context-dependent treatment of risk in the voluntary NIST AI RMF and Generative AI Profile [64, 65]; its scoring scheme is an author-proposed construction and does not determine regulatory risk classification.

4.3 RQ1: vulnerabilities and attack surfaces

The reviewed studies identify vulnerabilities across training and adaptation data, model assets and deployment configurations, instruction interfaces, external context and tools,

multimodal fusion, and agentic workflows. We use *attack surface* for the lifecycle location of adversarial influence, *mechanism* for how the attack operates, *modality* for the delivery channel, and *application domain* for where harm occurs. Thus, prompt injection is a mechanism, direct or indirect access characterises its delivery, multimodal identifies a channel, and healthcare identifies a consequence context. These consistent terms follow Tables 7 and 10.

4.3.1 Text-based jailbreaks and alignment limits

Text-based jailbreaks form a recurrent vulnerability class in the retained corpus. As one study-specific illustration, an analysis of 448 real-world jailbreak prompts classifies attack patterns into Disguised Intent, Role-Play, Structured Response, Virtual AI Simulation, and Hybrid categories, showing how attackers exploit instruction hierarchy, conversational context, and role simulation to bypass guardrails [100]. This result is not treated as a corpus-wide prevalence estimate; it illustrates that adversaries can manipulate context and meta-instructions rather than rely only on direct harmful requests.

Theoretical work further suggests that current alignment methods can reduce but may not fully eliminate jailbreak opportunities. The E-RLHF approach expands the set of safe outputs by identifying inputs that trigger harmful response normalisation and mapping them toward normative safe responses [82]. The relevant implication is not that alignment is ineffective, but that alignment alone should not be treated as a complete security boundary. Continuous red-teaming, layered detection, and post-deployment monitoring remain necessary.

Figure 16 illustrates a four-stage attack–defence cycle for multimodal AI systems. It begins with baseline attacks using direct jailbreaks and intent-disguise techniques, followed by primary defences based on system alignment and input/output guardrails. Attackers may then adapt through cross-modal and deployment-induced exploits, including visual prompt injections, universal image or text suffixes, and post-training compression attacks. The final counter-defence stage introduces multimodal verification, realignment, self-rationalisation, cross-modal consistency checks, visual screening, and post-compression re-benchmarking to strengthen the system's robustness against evolving adversarial strategies.

4.3.2 Prompt injection, indirect injection, and hidden external instructions

Applying the definitions in Sect. 2, the cross-study evidence shows that injection risk extends from user-controlled prompts to webpages, documents, retrieval sources, memory, and tool outputs. Hidden-link, document-injection, peer-review, and web-agent studies independently show that untrusted content can cross the instruction–data boundary and influence a model-enabled workflow [59, 83, 91, 108]. The resulting information-security implication is system-level: browsing, retrieval, code execution, and tool use must be included in the threat model rather than treating the prompt window as the complete attack surface.

4.3.3 Deployment-induced safety degradation

Deployment choices can alter safety behaviour even when the model family remains unchanged. In one study-specific quantisation experiment on Llama3-8B-Instruct, precision below 4 bits increased jailbreak success, while Korean prompts and past-tense formulations produced reported ASR increases of 25.5% and 39.3%, respectively [49]. These conditional values are not pooled with other studies. They motivate repeating the complete safety evaluation after compression, quantisation, pruning, distillation, or another deployment optimisation because a full-precision laboratory result may not transfer to the production configuration.

4.3.4 Multimodal and vision–language attacks

Multimodal models introduce additional attack channels because adversarial instructions can be hidden in visual or cross-modal inputs. In one study-specific mind-map experiment, malicious directives represented as visual nodes produced 90–100% attack success in the tested GPT-4o and Gemini settings [50]. The value is conditional on that attack, model version, and judgement protocol; its broader significance is that a text-only filter cannot inspect a malicious visual signal treated as task-relevant evidence.

Across additional vision–language studies, adversarial images transfer from public encoders to proprietary systems, and both white-box visual–text optimisation and black-box universal suffixes achieve high study-specific ASR under different access assumptions [29, 45, 87, 106]. Because the models, access levels, budgets, and judges differ, these values are not ranked or pooled. The convergent mechanism-level finding is that multimodal security requires input-channel isolation, visual-aware filtering, and cross-modal consistency checks rather than text-only safety screening.

4.3.5 Domain-specific consequences and validation requirements

Application context changes the consequence and acceptable threshold of a vulnerability; it is not coded as an attack mechanism. In one medical prompting study, harmful output rose from near zero to more than 90% for anti-vaccination content and from approximately 1% to more than 80% for unsafe drug-interaction responses in the tested setting [97].

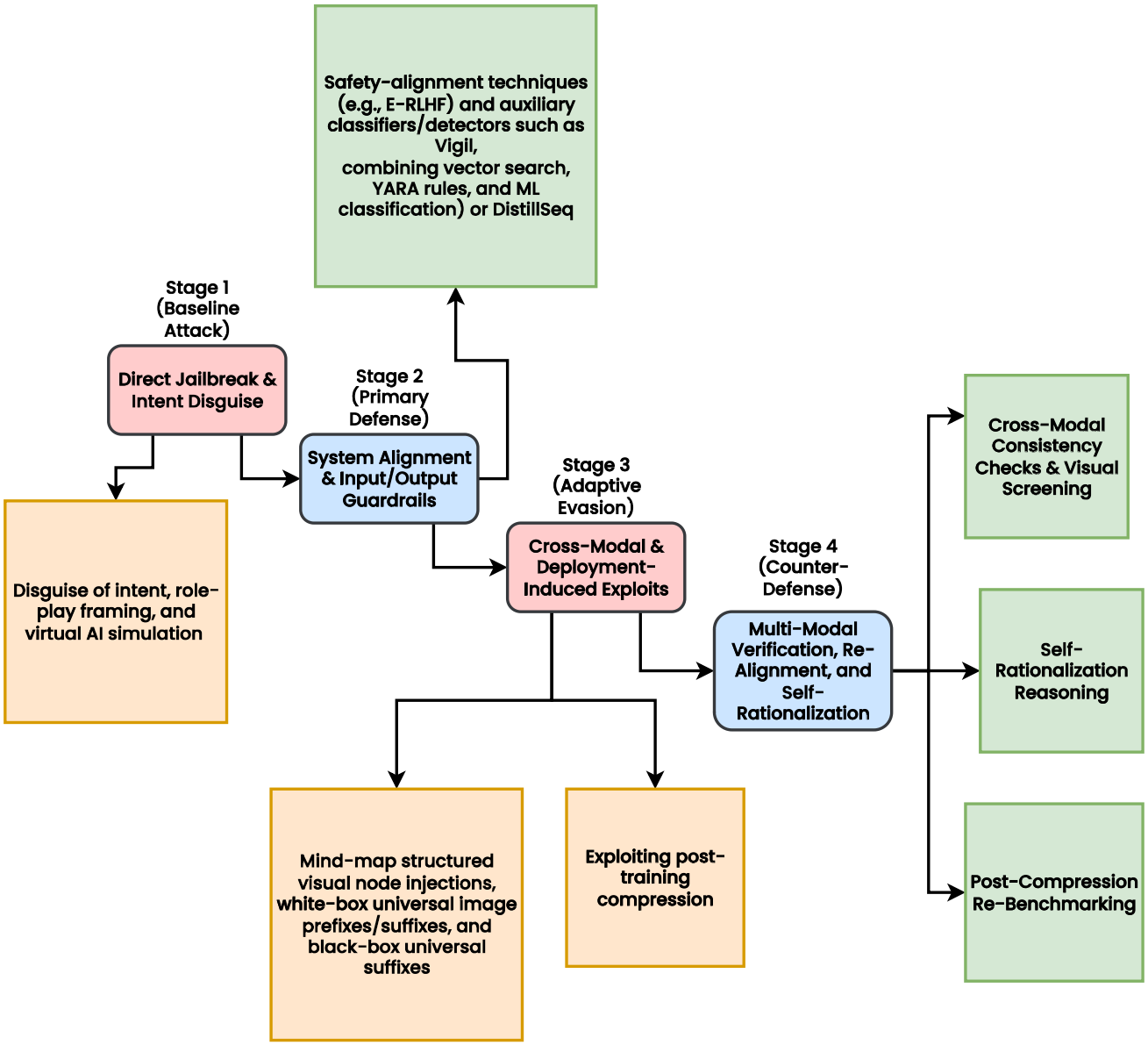


Fig. 16 Four-stage attack–defence cycle showing the progression from baseline jailbreak and intent-disguise attacks to alignment-based defences, adaptive multimodal evasion strategies, and counter-defence mechanisms for improving multimodal AI robustness

These study-specific values illustrate why clinical evaluation requires adversarial cases, expert judgement, and harm-based release criteria. In software security, sensitivity to class imbalance and complex code patterns similarly means that improved ROC-AUC or F1 in one fine-tuning study does not remove the need for adversarial and distribution-shift testing before code-auditing deployment [78].

4.4 RQ2: red-teaming and adversarial evaluation methods

The reviewed red-teaming literature uses manual, automated, benchmark-based, black-box, white-box, transfer-based, and multimodal testing methods. The central methodological trend is a move from ad hoc prompt testing toward structured adversarial evaluation. Structured jailbreak taxonomies support repeatability by defining prompt families, attack mechanisms, and evaluation outcomes [100]. DistillSeq shows how compact classifiers can support red-team prompt selection and prioritise high-risk cases, thereby improving coverage of adversarial testing [95]. These approaches indicate that red-teaming is increasingly treated as an engineering process rather than a purely exploratory exercise.

Black-box and white-box methods provide complementary views of model risk. Black-box attacks are realistic for external adversaries because they do not assume gradient

Table 12 Comparison of academic red-teaming and industrial frontier-safety pipelines

Pipeline	Scope and trigger	Evaluation evidence	Mitigation and decision process	Principal limitation
Reviewed academic practice	Study-defined model, task, attack surface, or benchmark; testing is usually triggered by a research question	Manual, automated, transfer, multilingual, multimodal, white-/black-box, and increasingly adaptive or agentic attacks; methods and results are peer reviewed	Defence may be evaluated experimentally, but deployment authority, incident response, and re-test triggers are often outside study scope	Heterogeneous versions, budgets, judges, metrics, and reporting reduce direct comparability
OpenAI Preparedness Framework v2 [67]	Covered frontier deployments, material capability changes, and significant agentic systems under tracked severe-risk categories	Threat models, scalable capability evaluations, deep dives, expert red-teaming, and third-party evidence where needed	Capabilities and Safeguards Reports support residual-risk review by a Safety Advisory Group, leadership decisions, and board oversight	Focuses severe frontier risk and necessarily discloses less procedural detail than an open academic benchmark
Google DeepMind Frontier Safety Framework v3.1 [27]	Frontier-model checkpoints before and after deployment, with Tracked and Critical Capability Levels	Risk identification, inherent-risk assessment, internal/external expertise, empirical testing, threat modelling, and mitigation stress-testing	Security and deployment mitigations lead to residual-risk assessment, risk acceptance, safety cases at critical levels, and post-market updates	Capability thresholds and acceptable residual risk retain context-dependent judgement and target severe-risk pathways
Anthropic Responsible Scaling Policy v3.4 [7]	Publicly deployed and qualifying high-risk internal models assessed against capability or usage thresholds	Capability and alignment evaluation, adversarial testing of safeguards, internal critique, external expert input, and incident evidence	A Frontier Safety Roadmap, periodic Risk Reports, external review, a Responsible Scaling Officer, and board/LTBT governance connect evidence to development and deployment decisions	The policy concentrates on catastrophic risk and does not replace product- or domain-specific safety evaluation

Table 13 Engineering comparison of representative defence families

Defence family	Applicable scope	Computational or operational overhead	Utility side effect	Limitation and residual risk
Prompt structuring and input screening [5, 70]	Interface-level direct injection and known jailbreak patterns; deployable without changing model weights	Rule/classifier execution and token overhead; PromptGuard reports less than 8% latency increase in its tested setting	Legitimate complex requests may be rejected or altered; threshold tuning affects over-refusal	Obfuscation, indirect context, non-text modalities, and adaptive attacks can bypass an interface-only control
Query–response and output safeguards [76, 90]	Inference-time screening of both the request and generated response; model-agnostic wrappers are possible	One or more detector/judge calls add latency, tokens, and service cost	A decontextualised judge may reject contextually legitimate content or miss nuance	Judge-model error, correlated model failures, policy drift, and adversarial attacks against the detector remain
Dedicated or ensemble injection classifiers [22, 77, 95]	High-throughput classification of prompt text and known attack families; useful at gateways and triage queues	Training, calibration, log curation, and periodic refresh; multi-agent variants consume additional context and calls	False positives reduce benign utility and language/domain transfer may be uneven	Strong static-benchmark F1 does not establish robustness to novel, adaptive, multilingual, or multimodal attacks
Adversarial training and robust adaptation [18, 73, 99]	Model-level resistance to represented attack families; suitable when weights or adapters can be updated	Additional training/optimisation, storage, validation, and regression-test cost	Catastrophic forgetting, clean-task degradation, or narrower response behaviour can occur	Protection may not transfer to unseen attacks, model updates, external tools, or changed modalities
Grounding, provenance, and context isolation [26, 59, 108]	RAG, document processing, web agents, and tool-enabled systems where untrusted content enters context	Retrieval/index maintenance, provenance checks, sanitisation, and possible extra verification calls	Strict source policies can lower recall or prevent useful external evidence from being used	Poisoned sources, hidden long-context instructions, stale indexes, and tool-chain propagation remain possible
Cross-model and human/domain verification [76, 79, 105]	High-consequence outputs requiring independent checking, domain thresholds, or human approval	Multiple inference calls and/or scarce expert time create latency, cost, and scaling constraints	Conservative escalation may slow workflows; reviewers may disagree or become fatigued	Correlated model errors, automation bias, insufficient expertise, and unreviewed edge cases leave residual risk

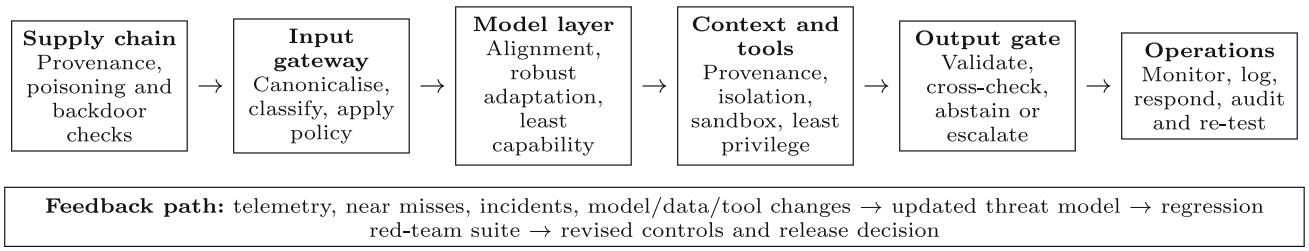


Fig. 17 Review-derived defence-in-depth architecture for foundation-model systems. No layer is treated as a complete trust boundary; evidence and incidents continuously update the next evaluation cycle

access or model internals, as shown by OpenSesame-style universal suffixes [45]. White-box attacks, including Universal Master Key optimisation, expose worst-case vulnerability under stronger access assumptions and are useful for stress-testing model and defence design [87]. Transfer attacks against VLMs further show that adversarial examples trained on smaller public models can affect larger or proprietary systems [106]. This transferability implies that security evaluation should not be limited to a single deployed model or a single vendor-specific interface.

Multimodal red-teaming is also becoming more important. Visual prompt injection and diffusion-based adversarial image generation demonstrate that attackers can exploit image channels, visual layouts, and cross-modal fusion mechanisms [29, 50]. In addition, quantisation-aware adversarial evaluation reveals that red-teaming should be repeated under realistic deployment configurations, not only under full-precision laboratory conditions [49]. Medical adversarial prompting shows that domain-specific red-teaming should include harmful advice, misinformation, unsafe clinical recommendations, and privacy-sensitive interaction patterns [97]. The evidence supports a multi-layer red-teaming protocol that specifies adversary access, attack modality, target failure mode, model configuration, benchmark source, and human or automated judgement procedure.

4.4.1 Academic and industrial red-teaming pipelines

Academic red-teaming and industrial deployment assurance have complementary strengths. The reviewed academic literature makes attack mechanisms and benchmark procedures inspectable, including autonomous, context-aware, adaptive, multilingual, and multimodal tests [30, 53, 92, 101, 105]. The industrial frameworks published by OpenAI, Google DeepMind, and Anthropic add formal capability triggers, deployment gates, residual-risk decisions, accountable governance, and post-deployment reassessment [7, 27, 67]. Table 12 compares these pipeline structures. The industrial documents are used as contextual practice sources and are not counted among the 94 peer-reviewed primary studies.

The comparison suggests a combined pipeline: publish an explicit threat model and versioned benchmark; use both scalable automated tests and expert deep dives; evaluate defence efficacy and benign utility; document residual risk; record an accountable release decision; and feed incidents and model changes into regression red-teaming. Academic studies can improve operational relevance by reporting release gates, owners, and re-test triggers, while industrial pipelines can improve external validity through reproducible benchmark detail and independent evaluation.

4.5 RQ3: defence mechanisms and mitigation evidence

The reviewed defence literature covers prompt-level safeguards, input and output filtering, dedicated detectors, adversarial training, robust fine-tuning, retrieval-augmented grounding, cross-model verification, domain-specific gatekeepers, and governance-oriented controls. The strongest recurring conclusion is that single-point defences are insufficient. Effective mitigation requires layered preventive, detective, containment, corrective, and monitoring controls. Table 13 therefore compares scope, overhead, side effects, and residual risk instead of pooling numerical improvements obtained under incompatible conditions.

All numerical examples in the following defence discussion are study-specific results under the cited paper's model, dataset, attack, and judgement conditions. They illustrate mechanisms and engineering trade-offs; they are not used to rank defences or estimate pooled effectiveness.

Except where a primary study reports a numerical value explicitly (for example, PromptGuard's tested latency increase), the overhead column identifies the additional component or workflow introduced by the defence and does not present an inferred runtime or cost. This distinction prevents architecture-implied expense from being misreported as experimental evidence.

Figure 17 integrates the reviewed controls into a deployment architecture. It begins before inference because poisoned data, backdoors, or compromised model assets cannot

be corrected reliably by an output filter alone. It then separates untrusted input and context from model instructions, constrains retrieval and tool privileges, validates outputs, and connects monitoring to incident response and re-testing. Recent layered query/response detection, structured injection resistance, adaptive optimisation, web-agent safeguards, and decontextualised output validation supply complementary checkpoints rather than substitutes [5, 73, 76, 90, 108].

4.5.1 Prompt and pipeline-level defences

Prompt-level defences remain attractive because they can be deployed without modifying model parameters. Prompt-G uses a meta-prompt that communicates safety intent and asks the model to examine and reject dangerous queries, thereby improving integrity at the interface layer [70]. RAG-based multimodal security systems with eligibility scoring report 98% accuracy in filtering injections and hallucinations, while cross-verification among multiple models improves downstream stability [26]. Related evaluations of RAG-style and prompt-level security controls show that pipeline orchestration, retrieval constraints, and intent verification can reduce hallucination and injection risk, although these controls also introduce new risks such as retrieval poisoning and source-quality dependence [74].

4.5.2 Dedicated detectors and gatekeeper models

Dedicated detectors act as upstream or downstream security components around the LLM. DistillSeq condenses model signals into a small classifier for selecting high-risk prompts and improving red-team coverage by 93% [95]. Vigil combines vector search, YARA rules, rule-based checks, and machine-learning classification to detect malicious prompts with 89% effectiveness [71]. Bio-linguistic analysis of jailbreak prompts shows that structural prompt features can distinguish malicious from benign prompts with high accuracy [47]. More advanced approaches use model self-rationalisation: Meticulous Thought Defender builds a reasoning tree and detects inconsistencies, improving accuracy by 1–3% over traditional detectors and by more than 6% over simple model self-checking [79]. ZDDR combines perplexity and fluency checks, restores accuracy through rephrasing, improves performance by 9%, and recovers 45% of lost performance in attacked settings [15].

4.5.3 Adversarial training and robust fine-tuning

Training-time defences attempt to improve robustness before deployment. RIFT reduces catastrophic forgetting and improves robustness by preserving mutual information between pretrained and fine-tuned models [18]. LoRA-Adv improves offensive-language detection by more than 6% while modifying only 0.06% of parameters, showing that parameter-efficient fine-tuning can improve robustness without full-model retraining [99]. Prompt-based adversarial example generation expands the adversarial training set and improves model resilience against crafted inputs [96]. Diffusion-based adversarial training extends this logic to multimodal models by generating high-quality adversarial images to improve robustness [29].

4.5.4 Domain-specific defences

Domain-specific safety strategies are often more effective than generic defences because they encode task constraints, domain vocabulary, and operational risk. Spiking Transformer-based multimodal emotion recognition systems show resilience to FGSM and PGD attacks [14]. Archias, a gatekeeper model for suppressing pathological responses in retail-oriented deployments, achieves a 19.6% improvement in effectiveness in legacy automotive applications [84]. KLAGE combines knowledge graphs, explainable AI, and LLM-generated explanations to detect IoT threats with more than 84% accuracy while improving interpretability for security analysts [10]. These examples indicate that security controls should be tailored to the domain asset, adversary model, and acceptable failure threshold.

Table 14 compares attack and detection families using a common set of characteristics: adversary access, target system, mechanism, evaluation evidence, response, and residual risk. This distinction matters because a high attack-success rate under white-box access and high detector accuracy on a static benchmark represent different security claims.

The comparison shows that attack realism and detector performance must be interpreted jointly. White-box optimisation provides an upper-bound stress test, whereas black-box, indirect, multilingual, multimodal, and agentic attacks more closely represent deployed systems. Detection studies should therefore report benign false positives, adaptive-attack performance, latency, model and policy version, and residual attack success after intervention.

4.5.5 Dynamic attack–defence co-evolution

Static evaluation assumes that an attack set is fixed while a defence is measured. The adaptive evidence instead supports a repeated game: at round t, an adversary selects $a_t = \mathcal{A}(H_{t-1}, d_t)$ after observing the deployed defence d_t and prior interaction history H_{t-1}; the defender then updates $d_{t+1} = \mathcal{D}(d_t, E_t)$ using evidence E_t about bypasses, benign utility, cost, and incidents. Autonomous jailbreak agents, context-aware red-teaming, self-evolving attacks, and adversarial-test-driven optimisation instantiate different parts of this loop [30, 53, 73, 92].

Table 14 Comparison of representative attack and detection methods in the reviewed literature

Method family	Threat model and target	Core mechanism	Evaluation evidence	Detection or mitigation	Limitation or residual risk
White-box token or embedding attacks [13, 28, 87]	Gradients, logits, or token scores; mainly open-weight LLMs/VLMs	Coordinate-gradient, spatial-momentum, relaxation, or joint visual–text optimisation	ASR, valid-token rate, convergence, and generation cost	Adversarial training, suffix screening, and robust fine-tuning	Strong worst-case test, but privileged access and closed-model transfer are uncertain
Autonomous, adaptive, and multi-agent jailbreaks [30, 32, 53, 92]	Query-only, multi-turn access to frontier models or customised applications	Reasoning agents plan, fuse, and evolve context-aware attacks	Nine- and 18-model tests; RedAgent tests 60 applications; ASR, turns, coverage, and cost	Continuous red-teaming, rate controls, state monitoring, and adaptive detectors	Results depend on model/policy version and judge criteria; repeated queries increase cost
Multilingual semantic attacks [81, 101]	Black-box prompts using script, translation, or culturally specific variation	Meaning is preserved while surface form changes to evade English-centred filters	Language-wise attack and detection outcomes across domestic and Indian multilingual models	Multilingual classifiers and culturally grounded adversarial suites	Dialect, transliteration, and code-mixing coverage remains incomplete
Indirect and cross-layer injection [59, 83, 108]	Attacker controls documents, webpages, retrieval, memory, or tool output	Hidden instructions cross the trusted-command/untrusted-content boundary	Six LLMs and 27 documents; 100 papers and 42,000 outputs; ASR, detection alert rate (DAR), and mitigation efficiency	Context isolation, provenance checks, sanitisation, and leakage-aware detection	Plausible long content can conceal attacks; tool-chain propagation remains possible
Multimodal injection [29, 50, 105]	Text or visual overlays; stronger settings use transferable or white-box perturbations	Instructions in images, diagrams, or video frames corrupt cross-modal fusion	ASR and accuracy degradation; surgical study tests four VLMs over 100 clips and eight tasks	Visual-aware filtering, cross-modal consistency checks, and adversarial training	Text-only controls cannot inspect covert or long-duration visual injections
Layered query/response detection [5, 90]	Inference-time wrapper; no target-model parameter access required	Input screening and structuring are combined with response reasoning or validation	Attack, harmful, adaptive, and benign benchmarks; safety, over-refusal, runtime, and F1	Chain-of-Detection and PromptGuard-style multi-stage controls	Additional inference cost; effectiveness depends on the downstream reasoning detector
Comparative and in-context classifiers [22, 77, 95]	Prompt-text classification using trained models or black-box multi-agent inference	Traditional ML, fine-tuned transformers, specialised detectors, or reusable error logs	More than 300,000 prompts in one benchmark; F1, false positives, and benign utility	Ensemble detection, calibrated thresholds, and continuously refreshed examples	Static-corpus accuracy may not survive adaptive shift; logs consume context and calls
Decontextualised output safeguard [76]	Secondary black-box safety model checks generated responses	A stateless judge evaluates output outside the possibly compromised conversation	Comparison with expert judgements and existing automated safeguards	Model-agnostic output validation and human escalation	Context removal can lose legitimate nuance; judge error and extra latency remain

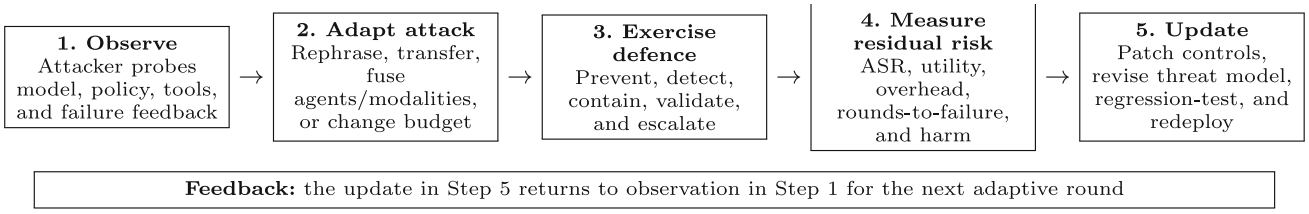


Fig. 18 Dynamic adversarial game linking attacker adaptation to defence updates and repeated residual-risk evaluation

Accordingly, a defence should be evaluated over multiple rounds and against a changing attacker, not only against the prompts used to design it. In addition to first-round ASR, the relevant outputs are worst-round and post-adaptation ASR, attacker query cost, rounds or time to first control failure, defender recovery time, retained benign utility, and marginal computational cost. Figure 18 also makes the authors' proposed engineering stopping rule explicit: release is justified only while residual risk remains within the domain's documented tolerance; a material model, data, tool, or policy change reopens the game.

4.6 RQ4: evaluation metrics and high-stakes validation

The reviewed evaluation practices combine task-performance metrics, attack-specific metrics, hallucination measures, expert ratings, and qualitative case analysis. Table 9 and Figs. 13–14 show that evaluation remains heterogeneous, which limits cross-study comparison. The framework recommends the reporting information that should accompany a result: model and system version, scale or configuration, adversary capability and budget, dataset and modality, attack intensity, benign utility, overhead, evaluator, uncertainty, quality tier, and residual risk.

To make the relationship between model choice and deployment context explicit, Table 15 maps representative foundation-model families to the principal application domains identified in the final 94-study corpus. The mapping is multi-label: a study may evaluate several models and contribute evidence to more than one application domain. It therefore describes model–application relationships rather than mutually exclusive study counts.

4.6.1 Clinical and biomedical use

The numerical values below are study-specific illustrations rather than pooled sector estimates. Their interpretation remains conditional on the cited dataset, model and version, task, evaluator, sample, and study-quality tier. Healthcare illustrates both the potential and the risk of LLM deployment. Consumer-facing medical education studies show that models can produce fluent patient-facing explanations, but correctness, readability, and hallucination rates vary across models [2]. In maxillomandibular fixation education, ChatGPT achieved the highest accuracy (88.5%) and the lowest hallucination rate (5.2%), while a Constitutional-AI model showed comparable hallucination control (6.0%) but produced responses at a postgraduate reading level [66]. This indicates that healthcare safety requires factual accuracy, interpretability, appropriate reading level, and alignment with professional standards.

Retrieval-augmented systems offer a practical way to reduce hallucination when the retrieval corpus is curated. EndoQ integrates dentistry textbooks and guidelines and receives higher expert scores (4.45/5) and professionalism ratings than generic LLMs [93]. NurRAG, designed for nursing education, improves precision by 43–73%, showing that domain-grounded retrieval can improve safety in clinical training contexts [91]. RAG does not eliminate all risk because retrieval quality, source integrity, and context selection can still affect output reliability.

4.6.2 Clinical prediction and decision support

Clinical prediction studies indicate that general-purpose LLMs do not automatically outperform domain-specific architectures. In mortality prediction, NSHNET, a note-specific hierarchical network, outperforms 34 LLMs by approximately 0.1 in AUC-ROC (0.88 versus 0.78), showing that long, noisy EHR notes remain difficult for general-purpose language models [54]. For ICU readmission prediction, narrative EHRs allow a lightweight Gemma2B model to reach an AUC of 0.90 and outperform the domain-oriented Apollo2B model [31]. A syncope-recognition system based on Italian and multilingual BERT models reaches an AUC of 0.98, further demonstrating the value of task-specific fine-tuning and domain-data alignment [51].

4.6.3 Information extraction, summarisation, and privacy

Healthcare workflows also depend on accurate information extraction and summarisation. Abstractive models can outperform extractive baselines on ROUGE, but models such as BigBird-Pegasus may introduce semantic distortion, which is a safety risk when clinicians rely on gener-

Table 15 Representative foundation-model families and their application domains in the reviewed literature

Application domain	Representative models or model families	Principal applications	Safety and robustness relationship	Representative studies
Healthcare and clinical medicine	Llama-3, Qwen, Vicuna, ChatGLM2, Gemma-2B, Apollo-2B, GatorTronGPT, GPT-4o, Gemini 1.5/2.5 Pro, GPT-o4-mini-high, and Qwen−2.5-VL	Clinical question answering, EHR analysis, mortality and readmission prediction, biomedical text generation, and surgical video decision support	Hallucination, privacy leakage, unsafe clinical advice, noisy-record sensitivity, and textual or visual injection can directly affect patient safety	[31, 54, 69, 97, 105]
Cybersecurity and AI-system safety	GPT-4/4o, Claude, Gemini, Llama, Vicuna, Qwen, DeepSeek, Grok, MiniGPT-4, LLaVA, WizardCoder, and dedicated guard models	Jailbreak testing, injection detection, software-vulnerability analysis, automated red-teaming, and web-agent protection	Model access, interaction mode, modality, application context, and configuration materially alter attack success and defence effectiveness	[30, 53, 78, 92, 108]
Business, finance, and organisational decision support	FinBERT, GPT-3/4, Codex, Vicuna, ChatGPT-4, Copilot, DeepSeek-V3, and Perplexity	Financial-text analysis, business-intelligence automation, document comparison, and decision support	Incorrect extraction, biased ranking, hidden document instructions, and non-transparent decisions create integrity and compliance risks	[33, 57, 59, 83]
Intelligent infrastructure and engineering	Time-LLM, TD-LLM, Industrial CLIP, spatiotemporal Transformer–LLM systems, GPT-4o-mini, GPT-4o, and o3-mini	Network-traffic forecasting, smart-grid recovery, industrial inspection, and collaborative engineering problem solving	Distribution shift, latency, numerical-error propagation, and a misleading agent can compromise operational or physical-system reliability	[25, 60, 89, 94]
Multilingual, low-resource, and social applications	SloBERTa, mBERT, IndicTrans, Mixtral-8x7B, Baichuan, ERNIE, GLM, Qwen, Spark, and Yi	Translation, code-mixed or regional-language analysis, harmful-content detection, and multilingual jailbreak or injection testing	Uneven language coverage permits semantic obfuscation and creates safety-filter blind spots across scripts, dialects, and low-resource languages	[62, 81, 85, 101]
Cross-domain deployment and red-teaming	DeepSeek-R1, Gemini 2.5 Flash, Grok 3 Mini, Qwen3, GPT-4o, Claude 4 Sonnet, Llama 3.1/4, and customised LLM applications	Autonomous adversarial testing, multi-turn attacks, context-aware application testing, and deployment assurance	Reasoning models can function as scalable attack agents, while application context and system integration determine the effective attack surface	[3, 30, 92]

Note: Models and studies may appear in more than one domain because many evaluations are cross-domain, multi-model, or multi-task. The table is descriptive and is not a set of mutually exclusive frequency counts.

ated summaries [12]. ZeroTuneBio NER and prescription-structuring pipelines show that carefully designed prompting and incremental-ratio strategies can achieve high entity-recognition accuracy without fine-tuning [37, 72]. Privacy-preserving NLP remains a distinct safety requirement: Azure's de-identification service reports an F1-score of 0.939 and outperforms several narrower GPT-4 few-shot alternatives in sensitive clinical contexts [43].

4.6.4 Foundation models, synthetic data, and domain adaptation

Biomedical foundation models can improve task performance but introduce additional validation requirements. GatorTronGPT, a 20B-parameter model trained on 277B tokens including 82B clinical tokens, achieves strong biomedical NLP results and can generate synthetic clinical text that improves smaller synthetic models such as GatorTronS [69]. Synthetic data may support privacy-preserving development, but it also requires validation for bias propagation, latent clinical-pattern distortion, and downstream safety. Domain adaptation appears similarly useful in finance and language-specific modelling: FinBERT improves financial text understanding [33], and SloBERTa shows the value of monolingual training for underrepresented languages [85]. These studies suggest that domain-specialised models can reduce some harmful failures, but domain adaptation should be accompanied by adversarial and distribution-shift testing.

Table 16 presents engineering recommendations derived from this review's evidence synthesis. The proposed assurance gates are not presented as statutory requirements. Some recommended practices may support activities relevant to applicable legal obligations, but neither adopting these gates nor satisfying their suggested criteria establishes legal compliance. The table replaces a ranking of isolated "best" accuracy or AUC values because a high task score cannot compensate for an untested injection path, absent human escalation, or an uncontrolled tool action. As an engineering recommendation, patient, physical, infrastructure, financial, or population-scale consequences warrant stronger independent validation, narrower authority, monitoring, and fail-safe behaviour than a reversible low-impact drafting task. The NIST AI RMF and Generative AI Profile provide voluntary context for consequence-sensitive risk management [64, 65]; the applicability of any EU AI Act obligations should be assessed separately [24].

4.6.5 Multilingual and low-resource mitigation loop

The multilingual evidence identifies partial mitigation solutions as well as shortages. Indian multilingual prompt-injection work directly evaluates language-aware detection [81]; Chinese semantic-obfuscation experiments expose transformations that a detector and adversarial training set must cover [101]; and monolingual or language-adapted models show how language-specific representation can improve low-resource task coverage [62, 85]. Table 17 closes part of the problem–solution loop by mapping this evidence to deployment actions and residual tests, but none establishes uniform safety across dialects, scripts, transliteration, code mixing, or changing cultural context.

The resulting loop is: discover a language-specific bypass, add it to a versioned multilingual suite, update the language-aware detector or robust adaptation, measure both residual ASR and benign over-refusal per language, obtain native-speaker review for consequential cases, and repeat after model or policy changes. This is more complete than merely identifying missing non-English benchmarks, while remaining explicit that current mitigation evidence is still uneven.

4.6.6 Security, social media, and other operational domains

Cybersecurity, code auditing, financial analysis, multilingual sentiment analysis, and content moderation are central to information-security deployment. In software vulnerability detection, WizardCoder-based models improve ROC-AUC and F1-score while handling severe class imbalance [78]. KLAGE combines knowledge graphs, LIME-style explainability, and LLM-generated narrative explanations to support IoT threat detection with high precision and analyst interpretability [10]. Web-crawling and meta-computing studies show that LLM-enabled systems are vulnerable to hidden-link poisoning and adversarial HTML/CSS manipulation, which can compromise retrieval, indexing, or tool-use pipelines [91]. These findings indicate that LLM security must be evaluated at the system level, not only at the model-output level.

Low-resource and social-media settings provide additional evidence that benchmark performance is not equivalent to deployment robustness. DepGPT achieves 97.96% accuracy and an F1-score of 0.98 for Bengali depression detection [17], and related studies address harmful-meme detection [75] and code-mixed sentiment analysis [41]. Informal language, adversarial rephrasing, sarcasm, rapidly evolving slang, and multimodal content can cause distributional shifts. These settings should therefore be treated as robustness and safety stress tests rather than merely as application domains.

4.7 RQ5: research gaps and future directions

The evidence-to-action synthesis in Fig. 15 identifies five connected gaps. First, benchmark and reporting fragmentation prevents dependable comparison: studies vary in prompts, model and policy versions, attack budgets, refusal rules, metrics, and human-judgement procedures. Compar-

Table 16 High-stakes assurance matrix: domain-specific constraints, cascading hazards, and review-derived deployment recommendations

Domain	Domain-specific safety constraints	Governance and potential legal considerations	Cascading hazard	Recommended assurance gate
Healthcare and clinical decision support [54, 97, 105]	Clinical validity, calibrated uncertainty, patient privacy, population shift, explainability appropriate to the user, and timely abstention	Professional accountability and clinical governance; assessment of applicable privacy and medical-device duties, and of any EU AI Act obligations for the relevant actor and system [24]	A hallucinated, injected, or visually manipulated recommendation may propagate into diagnosis, treatment, or delayed care	Prospective domain validation; adversarial and subgroup testing; clinician verification for consequential outputs; provenance, abstention, audit logging, and a safe fallback
Cybersecurity and software engineering [78, 92, 108]	Secret protection, correct code and tool execution, least privilege, adversarial input handling, reversibility, and low false reassurance	Secure-development controls, access accountability, evidence preservation, vulnerability and incident-response processes	Generated exploit code, compromised analysis, or an over-privileged agent can spread across repositories, services, or customer systems	Sandboxed execution; scoped credentials; independent code/security review; attack-chain red-teaming; rollback, telemetry, and incident escalation
Intelligent and critical infrastructure [25, 89]	Real-time latency, availability, state-estimation integrity, deterministic fail-safe behaviour, and robustness to sensor or communication faults	Critical-infrastructure governance, change control, operator accountability, and safety engineering; separate assessment of any applicable EU AI Act incident-reporting duties [24]	An erroneous recommendation or autonomous action can cascade through control loops, service availability, equipment, markets, or the physical environment	Digital-twin or simulation stress tests; hardware-/operator-in-the-loop validation; bounded authority; fail-safe state; manual override; staged rollout and continuous monitoring
Business, finance, and organisational decision support [33, 59, 83]	Confidentiality, source traceability, non-manipulable ranking or extraction, explainability, fairness, and reliable human approval	Assessment of applicable financial, consumer, employment, data-protection, and record-keeping duties; any EU AI Act obligations depend on the relevant actor and use case [24]	Hidden document instructions or fabricated evidence can influence credit, investment, hiring, procurement, or enterprise decisions at scale	Verified sources; separation of instructions from content; approval thresholds and segregation of duties; bias/robustness tests; immutable decision records and appeal routes

Note: The assurance gates are engineering recommendations derived from this review. They are not presented as statutory requirements or as a conformity-assessment procedure. Any applicable legal obligations require a separate assessment, and adopting these recommendations does not establish compliance

Table 17 Multilingual safety problems, available mitigations, and unresolved validation needs

Observed problem	Existing or evidence-supported mitigation	Deployment implementation	Residual gap and required test
Semantic obfuscation preserves harmful meaning while changing the Chinese surface form [101]	Language-specific adversarial examples and semantic rather than keyword-only detection; include obfuscation families in robust adaptation	Normalise script where appropriate, retain the original input, compare semantic and lexical detector signals, and escalate disagreement	Evaluate regional vocabulary, new obfuscations, false rejection, transfer across model/policy versions, and adaptive reformulation
Injection and detection behaviour varies across Indian languages and model families [81]	Multilingual injection classifiers and language-wise thresholds rather than an English detector applied unchanged	Route by identified language/script, log performance per language, and refresh detectors with local adversarial examples	Test code mixing, transliteration, dialects, low-sample languages, calibration, and benign over-refusal separately
Low-resource representations and cultural context are sparsely covered [62, 85]	Monolingual or regionally adapted encoders, curated local corpora, and community/domain-expert annotation can improve linguistic coverage	Pair automated evaluation with native-speaker review and culturally specific harm taxonomies	Report annotator provenance and disagreement; avoid treating translation-equivalent prompts as culturally equivalent harms
Cross-language transfer can expose a filter that appears robust in one language	*Authors' synthesis:* translation/round-trip comparison and ensemble disagreement are proposed as an additional detection layer, not a sole defence	Compare original-language, canonicalised, and translated safety decisions while preserving privacy and latency budgets	Measure translation distortion, unequal service quality, additional cost, and attacker adaptation to the ensemble

ative prompt-injection and indirect-injection studies show the value of common benchmarks, but also demonstrate that results remain conditional on the threat model and context [22, 59]. Future work should publish versioned protocols that jointly report attack success, retained utility, over-refusal, latency, cost, and residual risk.

Second, multilingual, low-resource, and cross-modal coverage remains uneven. Chinese semantic obfuscation and Indian multilingual injection studies show language-dependent bypass and detection behaviour [81, 101]; surgical vision–language injection demonstrates that textual conclusions cannot be transferred automatically to visual clinical workflows [105]. The corresponding priority is culturally grounded, per-language and per-modality evaluation with domain-expert harm thresholds.

Third, static and short-term testing understates adaptive and agentic threats. Autonomous jailbreak agents, context-aware red-teaming, self-evolving attacks, and multi-agent prompt fusion adapt across turns or application context [30, 32, 53, 92]. Future evaluations should therefore measure post-adaptation attack success, query cost, time-to-defence-failure, transfer after model updates, and performance under sustained adversarial pressure.

Fourth, model-only evaluation misses cross-layer propagation through documents, retrieval, memory, tools, and interacting agents [83, 89, 108]. The corresponding research priority is a deployment-level test harness that checks provenance, privilege separation, context isolation, cross-modal consistency, and output validation as a single defence-in-depth chain.

Fifth, governance remains weakly coupled to technical evidence. Safety controls should be linked to audit logging, access control, incident response, post-deployment monitoring, documentation, and human escalation. Longitudinal studies should report not only whether a defence passes a benchmark, but also how detected events are triaged, remediated, audited, and used to update the next red-teaming cycle.

Table 18 converts these themes into a staged programme with deliverables and validation gates. The horizons express dependency rather than a forecast: comparable reporting and datasets are prerequisites for credible adaptive testbeds, which in turn are prerequisites for longitudinal assurance and standardisation.

The near-term stage is supported by current comparative, multilingual, and indirect-injection studies [22, 59, 81, 101]. The medium-term stage builds on autonomous and self-evolving attackers and adaptive defences [30, 53, 73, 92]. The long-term stage addresses the evidence absent from short benchmark studies: operational incidents, defence ageing, organisational response, and independently reviewable residual-risk decisions.

Table 18 Staged and actionable future-research roadmap

Stage	Priority actions	Proposed deliverables	Proposed validation milestones
Near term (0–12 months)	Adopt the reporting fields in Table 9; publish model-, policy-, data-, judge-, and attack-budget versions; run independent replications; expand multilingual, multimodal, agent, poisoning/backdoor, and model-extraction coverage	Machine-readable experiment cards, shared code and prompts where safe, per-language/per-modality results, benign-utility baselines, and explicit Good/Fair evidence tags	A result is reproducible by an independent team and can be compared without hiding material differences in model, task, access, or attack intensity
Medium term (1–3 years)	Build multi-round adaptive attack–defence testbeds; test complete RAG/tool/agent systems; benchmark overhead and side effects; conduct domain-expert and independent high-stakes pilots	Versioned regression suites, defence-in-depth reference implementations, safety cases, cost/latency profiles, and health-care/cyber/infrastructure assurance protocols	Residual risk remains within a documented domain threshold after adaptive re-testing, while utility, latency, and human workload remain acceptable
Long term (beyond 3 years)	Establish longitudinal incident and near-miss repositories; study model/data/tool drift and attack–defence equilibria; harmonise technical evidence with audit, certification, procurement, and incident governance	Privacy-preserving incident standards, cross-vendor evaluation interfaces, continuously updated safety cases, and evidence-based accreditation or conformity-assessment methods	Independent monitoring shows that risk remains controlled across releases and real deployments, with traceable corrective action when assumptions or controls fail

Note: The stages, deliverables, and validation milestones are proposed research priorities. They are not presented as regulatory deadlines or statutory requirements

4.8 Discussion: technical–governance safety assurance

The evidence indicates that foundation-model security is a system and lifecycle property. Alignment at the base model is necessary but insufficient because risk emerges from training and adaptation data, model assets, prompts, retrieval, modalities, tools, permissions, interacting agents, and organisational decisions. We therefore recommend identifying the protected asset, relevant confidentiality/integrity/availability/privacy/safety property, adversary capability, deployment consequence, and owner of the residual-risk decision when defining the security objective.

The NIST AI RMF provides voluntary guidance through the continuous Govern–Map–Measure–Manage functions, while its Generative AI Profile emphasises governance, content provenance, pre-deployment testing, and incident disclosure [64, 65]. These documents do not themselves impose statutory obligations. The EU AI Act establishes legal obligations for actors within its scope; the relevant duties depend on the actor's role, the system or model concerned, the applicable provisions, and their application dates. Depending on those provisions, obligations may concern documentation, risk management, adversarial testing, cybersecurity, post-market monitoring, or serious-incident reporting. Article 55 specifies additional obligations for providers of general-purpose AI models with systemic risk [24]. Figure 19 maps the review's proposed engineering practices to these governance concepts. The mapping is descriptive: it indicates how evidence from the recommended practices may support relevant risk-management or compliance activities, without asserting that the proposed practices are legally required or that adopting them establishes compliance. These contextual sources do not contribute to the 94-study descriptive counts.

Table 19 translates this closed loop into concrete organisational docking barriers, engineering mechanisms, and auditable outputs.

This closed loop resolves the superficial separation of safeguards and governance: red-teaming supplies measurable evidence, defence-in-depth maps evidence to controls, governance assigns authority and thresholds, and deployment telemetry tests whether the original assumptions remain valid. The principal practical recommendation is traceability. We recommend linking each benchmark result to the exact system version, a threat and harm scenario, the implemented control, its operational cost and side effects, an accountable release decision, and post-deployment evidence.

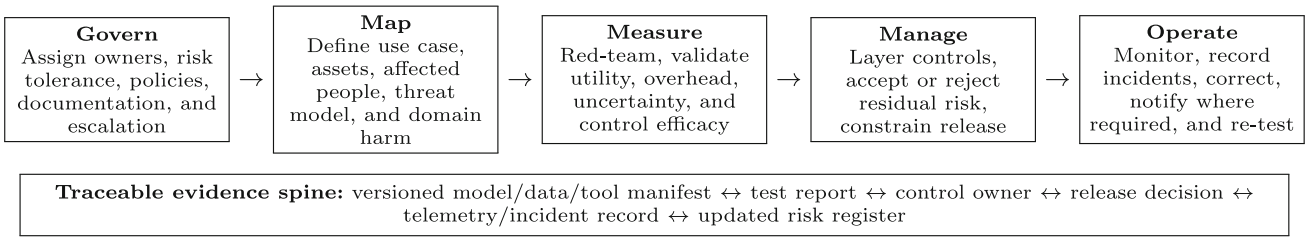


Fig. 19 Proposed technical–governance assurance cycle informed by the review and the voluntary NIST AI RMF. The mapping is descriptive; the engineering recommendations are not presented as legal requirements, and adopting them does not establish compliance with the EU AI Act or other legislation

Table 19 Engineering barriers at the technical–governance interface and practical docking mechanisms

Integration barrier	Why technical evidence fails to dock	Practical mechanism	Auditable output
Metric-to-risk mismatch	ASR, F1, or hallucination rates are reported without a business/domain consequence, risk owner, or tolerance	Link every test case to an asset, harm scenario, severity rubric, control, and acceptance threshold	Threat/control matrix and signed residual-risk decision
Version and configuration drift	A report becomes stale after a model, policy, retrieval corpus, tool, precision, or prompt change	Maintain a machine-readable configuration manifest and automatic re-test triggers for material changes	Versioned evidence package and regression history
Fragmented ownership	ML, security, product, legal/compliance, and domain experts operate separate queues and vocabularies	Assign a named control owner and decision owner; define escalation, exception, and incident responsibilities	Responsibility matrix, approvals, exceptions, and closure records
Telemetry versus privacy	Rich prompts and traces aid detection but may expose personal, proprietary, or security-sensitive data	Apply data minimisation, access separation, retention limits, secure aggregation, and privacy review to safety telemetry	Data-flow map, access log, retention schedule, and privacy/security assessment
Third-party opacity	A downstream deployer may lack model internals, training data, or timely notice of vendor changes	Specify supplier documentation, version/change notification, evaluation rights, and incident channels in supplier agreements; implement compensating wrapper controls locally	Supplier assurance record, service obligations, and local validation results
Release cadence versus review latency	Manual governance cannot keep pace with frequent model and application updates	Automate evidence collection and policy gates while reserving human approval for high-impact exceptions and residual risk	Continuous-assurance dashboard, gated release record, and time-bounded exception

Note: The mechanisms and auditable outputs are engineering and organisational practices recommended by the authors. They are not presented as statutory requirements, and implementing them does not establish legal compliance

5 Conclusion

This SLR reviewed peer-reviewed primary research published from January 2021 to July 2026. Of the 101 studies entering quality assessment, seven poor-quality studies were excluded, producing a final synthesis of 94 studies. The quantitative coding shows that 48 studies (51.1%) investigated multiple vulnerability categories and 81 (86.2%) used multiple red-teaming methods. In contrast, only 38 studies reported multiple mitigation strategies, while 15 reported no mitigation strategy. Thus, the breadth of adversarial discovery has expanded more rapidly than comparable evidence for layered and deployment-ready defences.

The principal analytical finding is that LLM safety failures are cross-layer phenomena. Instruction attacks can be combined with untrusted external context, deployment-induced

changes, visual inputs, memory, tools, and interacting agents. Recent evidence on autonomous and context-aware jailbreak agents [30, 92], self-evolving attacks [53], indirect injection in documents and web-agent contexts [59, 83, 108], multilingual bypass and detection [81, 101], and injection in surgical vision–language systems [105] demonstrates why static, text-only, and single-model tests can understate deployment risk.

The taxonomy retains data poisoning, backdoors, and model stealing for completeness, but the empirical coverage of these mechanisms is limited by the search strategy. The 94-study corpus is not fully representative of these research areas, and their sparse representation should not be interpreted as evidence of low real-world prevalence or severity.

These findings support a unified assessment recommendation. ASR, hallucination rate, robust accuracy, and detector performance should not be pooled without aligned model and policy versions, task and dataset, attacker access and budget, modality, judge, and defence setting. Each evaluation should combine adversarial effectiveness with benign utility, over-refusal, latency, computational or human cost, uncertainty, study-quality tier, and residual risk. Recent layered, comparative, and output-validation approaches provide useful foundations, but none removes the need for adaptive re-testing and system-level assurance [5, 22, 76, 90].

For multilingual and high-stakes applications, we recommend a complete problem-solution loop: language- and domain-aware mitigation, expert validation, explicit harm thresholds, and monitoring after deployment. The staged roadmap progresses from reproducible measurement, through adaptive defence-in-depth testbeds, to longitudinal technical-governance assurance. Trustworthy deployment should therefore be treated as a continuous, accountable process connecting versioned threat modelling, repeated red-teaming, layered controls, domain-specific release gates, incident evidence, and governance rather than as a one-time model-level score. As clarified in Sect. 2.3, the proposed practices and release gates are engineering recommendations derived from the evidence synthesis. Their adoption does not establish legal compliance; any applicable statutory obligations arise independently from the relevant legislation.

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Declarations

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Ethical approval As this is a survey paper and all data were obtained from the publicly available domain, no ethical approval or permission was required for the conduct of this study.

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